

Human review packet: GKP26LinearRepresentationHypothesis

Generated from byte-pinned source excerpts, the expanded PaperInterface specifications, and hash-bound raw-source-to-Spec screening records.

Paper: GKP26LinearRepresentationHypothesis
Paper id: GKP26LinearRepresentationHypothesis
Source version: arXiv v1, 2026
Source record: audit/paper_statement_map.json
Rows: 11
Generated: 2026-09-06
Source URL: [official source](#)

How to use this packet

For each source claim, compare the exact source excerpts with the expanded PaperInterface specification. Mark up this PDF or fill the blank reviewer box in the TeX source; return the annotated file for reconciliation into the paper-local human-review log.

Open the interactive dashboard

The interactive dashboard is an optional alternative to using this PDF; it presents the same review surface rather than an additional review requirement. After forking and cloning (or pulling) AppliedModelingLib, run the following from the repository root. The cache command is needed only the first time in a new clone.

```
cd <your-repository-checkout>
lake exe cache get
python3 scripts/review_dashboard.py --paper GKP26LinearRepresentationHypothesis --serve
```

Open <http://127.0.0.1:8765/> in a browser and keep that terminal running while reviewing. The dashboard records saved annotations in this paper's local reviewer-owned review record.

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 - `paper_compressed_sensing_external_basis_pursuitSpec`
 - `paper_geometry1_source_constructed_rateSpec`
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Material library prerequisites

AppliedModelingLib.FairCoin.fairMeasure

Source locator: cited publication, lines 412-418

Verbatim paper-source connection

```
\begin{lemma}[Incoherence of Random Matrices]\label{prop:incoherent}
  Let  $C \in \mathbb{R}^{d \times m}$  be a matrix with i.i.d. scaled Rademacher entries  $\pm \frac{1}{\sqrt{d}}$ . For
\begin{equation}
  d = O\left(\frac{\log m + \log \frac{1}{\delta}}{\mu^2}\right),
\end{equation}
 $C$  is  $\mu$ -incoherent with probability at least  $1 - \delta$ .
\end{lemma}
```

Lean-expanded library semantic target

```
type:
MeasureTheory.Measure Bool

value:
(PMF.bernoulli (1 / 2) AppliedModelingLib.FairCoin.fairMeasure._proof_2).toMeasure
```

Exact Lean library declaration

```
noncomputable def fairMeasure : Measure Bool :=
  ((PMF.bernoulli (1 / 2 : NNReal) (by norm_num))).toMeasure : Measure Bool)
```

Recorded source-to-library screening Verdict: matches

Reason: The pinned random-matrix proof uses independent uniform signs in $\{-1,1\}$; fairMeasure is the exact finite sign law used to express one such Rademacher coordinate.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.FairCoin.productMeasure

Source locator: cited publication, lines 412-418

Verbatim paper-source connection

```
\begin{lemma}[Incoherence of Random Matrices]\label{prop:incoherent}
  Let  $C \in \mathbb{R}^{d \times m}$  be a matrix with i.i.d. scaled Rademacher entries  $\pm \frac{1}{\sqrt{d}}$ . For
  \begin{equation}
    d = O\left(\frac{\log m + \log \frac{1}{\delta}}{\mu^2}\right),
  \end{equation}
   $C$  is  $\mu$ -incoherent with probability at least  $1 - \delta$ .
\end{lemma}
```

Lean-expanded library semantic target

```
type:
(ι : Type u_1) → MeasureTheory.Measure (ι → Bool)

value:
fun (ι : Type u_1) => MeasureTheory.Measure.infinitePi fun (x : ι) => AppliedModelingLib.FairCoin.fairMeasure
```

Exact Lean library declaration

```
noncomputable def productMeasure (ι : Type*) : Measure (ι → Bool) :=
  Measure.infinitePi (fun _ : ι => fairMeasure)
```

Recorded source-to-library screening Verdict: matches

Reason: The pinned proof requires independent Rademacher entries across rows and columns; productMeasure is the finite product construction of that stated joint law.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.FiniteDimensionalNorms.l1

Source locator: cited publication, lines 184-199

Verbatim paper-source connection

Let us recall a classical result in compressed sensing \citep{candes2006near, candes2006robust, donoho2006compressed}, which exactly deals with the setting in which features are represented linearly but accessed non-linearly, answering Q1.

```
\begin{theorem}[Compressed Sensing]\label{thm:compressed-sensing}
  There exists a matrix  $A$  in  $\mathbb{R}^{d \times m}$  with  $d = O(k \log \frac{m}{k})$ 
  such that for all  $k$ -sparse  $z$  in  $\mathbb{R}^m$ ,
  \begin{equation}
    g(x) := \argmin_{z' \in \mathbb{R}^m: x=Az'} \|z'\|_1
  \end{equation}
  satisfies
  \begin{equation}
    g(Az) = z.
  \end{equation}
\end{theorem}
```

This shows that when

```
\begin{equation}
  d = O(k \log \frac{m}{k}),
\end{equation}
```

Settled source reading Compressed sensing: the checked construction restricts the unstated sparse range to $1 \leq k$ and $2k \leq m$.

Lean-expanded library semantic target

```
type:
  {ι : Type u_1} → [Fintype ι] → (ι → ℝ) → ℝ

value:
  fun {ι : Type u_1} [Fintype ι] (x : ι → ℝ) => ∑ i : ι, |x i|
```

Exact Lean library declaration

```
/-- Finite-coordinate `L1` quantity, written as a sum of absolute values. -/
def l1 {ι : Type*} [Fintype ι] (x : ι → ℝ) : ℝ :=
  ∑ i : ι, |x i|
```

Recorded source-to-library screening **Verdict:** matches

Reason: The compressed-sensing source minimizes the finite-dimensional l1 norm of z' . The Lean declaration is exactly the sum over coordinates of $|x_i|$, which is that l1 objective on $\mathbb{R}^m$.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.FiniteDimensionalNorms.l2Sq

Source locator: cited publication, lines 638-655

Verbatim paper-source connection

```
\begin{proposition}\label{prop:geometry-2}
Consider any constants  $\epsilon > 0$ ,  $\gamma \geq 1$ . Then if  $\|a_i\|, \|b_i\| \leq \gamma$  for all  $i \in [m]$  and
\begin{equation}
\|B^T A z - z\|_\infty < \epsilon
\end{equation}
for all  $k$ -sparse  $z \in [-1, 1]^m$ , it holds that
\begin{enumerate}
\item[(i)]  $\angle \frac{a_i}{\|a_i\|}, \frac{b_j}{\|b_j\|} \geq \frac{1-\epsilon}{\gamma^2}$  for all  $i \in [m]$ ,
\item[(ii)]  $\angle \frac{a_i}{\|a_i\|}, \frac{a_j}{\|a_j\|} \leq \frac{\epsilon \gamma^2}{1-\epsilon} + \sqrt{1 - \frac{(1-\epsilon)^2}{\gamma^4}}$  for all  $i \neq j$ ,
\item[(iii)]  $\angle \frac{b_i}{\|b_i\|}, \frac{b_j}{\|b_j\|} \leq \frac{\epsilon}{1-\epsilon} + \sqrt{1 - \frac{(1-\epsilon)^2}{\gamma^4}}$  for all  $i \neq j$ .
\end{enumerate}
\end{proposition}

\begin{proof}
We must have that  $\angle a_i, b_i \geq 1 - \epsilon$  since  $\|B^T A e_i - e_i\|_\infty \leq \epsilon$ . It follows directly that
\begin{equation}
\angle a_i, b_i > 1 - \epsilon \implies \left| \angle \frac{a_i}{\|a_i\|}, \frac{b_i}{\|b_i\|} \right| \geq \frac{1-\epsilon}{\gamma^2},
\end{equation}
```

Settled source reading Geometry Proposition 2: $\epsilon > 0$ becomes $0 < \epsilon < 1$; the first denominator in parts (ii)/(iii) changes from $1 - \epsilon$ to $(1 - \epsilon)^2$, giving a weaker finite bound.

Lean-expanded library semantic target

```
type:
{ι : Type u_1} → [Fintype ι] → (ι → ℝ) → ℝ

value:
fun {ι : Type u_1} [Fintype ι] (x : ι → ℝ) => ∑ i : ι, x i ^ 2
```

Exact Lean library declaration

```
/-- Squared finite-coordinate `L2` quantity. -/
def l2Sq {ι : Type*} [Fintype ι] (x : ι → ℝ) : ℝ :=
  ∑ i : ι, x i ^ 2
```

Recorded source-to-library screening Verdict: matches

Reason: The Proposition 2 source uses squared Euclidean column norms through γ^2 and normalized inner products. The Lean value $\sum_i x_i^2$ is exactly the squared Euclidean norm on the finite coordinate space.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

```

\begin{proposition}\label{prop:geometry-1}
Consider constants  $\epsilon, \delta \in (0,1)$ . Then for  $d = O_{\epsilon, \delta}(k^2 \log m)$ , there exists  $A, B \in \mathbb{R}^{d \times m}$  such that
\begin{equation}
\|B^T A z - z\|_{\infty} < \epsilon
\end{equation}
for all  $k$ -sparse  $z \in [-1,1]^m$ , and where
\begin{enumerate}
\item(i)  $|\langle \frac{a_i}{\|a_i\|}, \frac{b_j}{\|b_j\|} \rangle| < \delta + o(1)$  for all  $i \in [m]$ ,
\item(ii)  $|\langle \frac{a_i}{\|a_i\|}, \frac{a_j}{\|a_j\|} \rangle| > 1 - \delta - o(1)$  for all  $i \neq j$ ,
\item(iii)  $|\langle \frac{b_i}{\|b_i\|}, \frac{b_j}{\|b_j\|} \rangle| > 1 - \delta - o(1)$  for all  $i \neq j$ .
\end{enumerate}
\end{proposition}

\begin{proof}
We give a construction. Let  $c_1, c_2, \dots, c_m, a^*, b^*$  be the columns of a  $\mu$ -incoherent matrix  $\mathbb{R}^{d \times m+2}$ . Then set
\begin{align}
a_i &= c_i + \lambda a^* \\
b_i &= c_i + \lambda b^*
\end{align}
for  $\lambda = \sqrt{\frac{1}{\delta}} - 1$ .
We begin by showing that for an appropriately chosen  $\mu$ ,  $\|B^T A z - z\|_{\infty} < \epsilon$  for all  $k$ -sparse  $z \in [-1,1]^m$ . Now let  $\hat{z} = B^T A z$  for any  $z \in [-1,1]^m$  with support  $T$ . Then
\begin{equation}
\hat{z}_i = \langle b_i, \sum_{j=1}^m z_j a_j \rangle = \langle b_i, a_i \rangle z_i + \sum_{j \in T^c} z_j \langle b_i, a_j \rangle.
\end{equation}
Therefore,
\begin{align}
|\hat{z}_i - z_i| &\leq |\langle b_i, a_i \rangle - 1| + \sum_{j \in T^c} |z_j \langle b_i, a_j \rangle| \\
&\leq |\langle b_i, a_i \rangle - 1| + \sum_{j \in T^c} |\langle b_i, a_j \rangle|,
\end{align}
where the last inequality follows from observing that  $z_j \in [-1,1]$  for all  $j \in [m]$ . Now observe that
\begin{align}
|z_i \langle b_i, a_i \rangle - 1| &= |z_i| |\langle c_i + \lambda b^*, c_i + \lambda a^* \rangle - 1| \\
&\leq |z_i| (|\langle c_i, \lambda b^* \rangle| + |\langle \lambda b^*, c_i \rangle| + |\langle \lambda b^*, \lambda a^* \rangle|) \\
&\leq |z_i| (2\lambda + \lambda^2 \mu) \tag{eq:apple-2} \\
&\leq |z_i| (1 + \lambda)^2 \mu,
\end{align}
where \tag{eq:apple-1} follows because  $|\langle c_i, c_i \rangle| = 1$  and \tag{eq:apple-2} follows from  $\mu$ -incoherence. Similarly,
\begin{align}
|\langle b_i, a_j \rangle| &= |\langle c_i + \lambda b^*, c_j + \lambda a^* \rangle| \\
&= |\langle c_i, c_j \rangle| + |\langle c_i, \lambda a^* \rangle| + |\langle \lambda b^*, c_j \rangle| + |\langle \lambda b^*, \lambda a^* \rangle| \\
&\leq (1 + 2\lambda + \lambda^2 \mu) = (1 + \lambda)^2 \mu.
\end{align}
This implies that
\begin{equation}
|z_i \langle b_i, a_i \rangle - 1| + \sum_{j \in T^c} |\langle b_i, a_j \rangle| \leq k (1 + \lambda)^2 \mu,
\end{equation}
where we use the fact that if  $|z_i| > 0$ , then  $i \in T$  and  $|T^c| \leq k-1$ . Therefore, it suffices to take  $\mu = \frac{\epsilon}{k(1+\lambda)^2}$  to ensure that  $\|B^T A z - z\|_{\infty} < \epsilon$ . From \tag{prop:incoherent}, there exists a  $\mu$ -incoherent matrix in  $\mathbb{R}^{d \times m+2}$  for
\begin{equation}
d = O\left(\frac{k^2(1+\lambda)^4}{\epsilon^2} \log(m+2)\right) = O\left(\frac{k^2(1+\lambda)^4}{\epsilon^2} \log m\right) = O_{\epsilon, \delta}(k^2 \log m).
\end{equation}

\paragraph{Establishing (i).} We now show that the construction satisfies (i). Observe that for all  $i \in [m]$ ,
\begin{align}
\left| \left\langle \frac{a_i}{\|a_i\|}, \frac{b_i}{\|b_i\|} \right\rangle \right| &= \left| \frac{\langle c_i + \lambda b^*, c_i + \lambda a^* \rangle}{\|c_i + \lambda b^*\| \|c_i + \lambda a^*\|} \right| \\
&\leq \frac{|\langle c_i, c_i \rangle| + |\langle c_i, \lambda a^* \rangle| + |\langle \lambda b^*, c_i \rangle| + |\langle \lambda b^*, \lambda a^* \rangle|}{\|c_i + \lambda b^*\| \|c_i + \lambda a^*\|} \\
&\leq \frac{1 + (2\lambda + \lambda^2 \mu)}{(1 + \lambda)^2} = \frac{1 + \frac{2\epsilon}{k(1+\lambda)^2}}{(1 + \lambda)^2} = \frac{1 + o(1)}{(1 + \lambda)^2} \tag{eq:dog}
\end{align}
Note that
\begin{align}
\langle a_i, a_i \rangle &= \langle c_i + \lambda a^*, c_i + \lambda a^* \rangle = \langle c_i, c_i \rangle + 2\langle c_i, \lambda a^* \rangle + \langle \lambda a^*, \lambda a^* \rangle \\
&= 1 + \lambda^2 - 2\lambda \frac{\epsilon}{k(1+\lambda)^2} = 1 + \lambda^2 - o(1).
\end{align}

```

```
type:
{ι : Type u_1} → [Fintype ι] → (ι → ℝ) → ℝ

value:
fun {ι : Type u_1} [Fintype ι] (x : ι → ℝ) => √(AppliedModelingLib.FiniteDimensionalNorms.l2Sq x)
```

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```
-- Finite-coordinate `L2` quantity, written as the square root of `l2Sq`. -/
def l2 {ι : Type*} [Fintype ι] (x : ι → ℝ) : ℝ :=
  Real.sqrt (l2Sq x)
```

Recorded source-to-library screening **Verdict:** matches

Reason: Geometry Proposition 1 uses Euclidean column norms. The Lean declaration takes the square root of l2Sq, whose displayed value is the finite sum of coordinate squares, exactly the Euclidean norm on $\mathbb{R}^d$.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Math.LinearCompressedSensing.InUnitBox

Source locator: cited publication, lines 205-209

Verbatim paper-source connection

Let $d(m,k,\epsilon)$ be the smallest choice of d such that there exists $A, B \in \mathbb{R}^{d \times m}$ such that

$$\|B^T A z - z\|_\infty < \epsilon$$

for all k -sparse $z \in [-1, 1]^m$. (Here, any bounded interval would suffice.)

[Next verbatim source excerpt]

Intuition and Proof Techniques

We begin by developing some geometric intuition for the problem. Let $a_1, \dots, a_m \in \mathbb{R}^d$ be the columns of $A \in \mathbb{R}^{d \times m}$ and $b_1, \dots, b_m \in \mathbb{R}^d$ be the columns of $B \in \mathbb{R}^{d \times m}$. For each $i \in [m]$, we recall that a_i is feature i 's **representation vector** and b_i is feature i 's **probe vector**. Now let $\hat{z} = B^T A z$. We are interested in when

$$\|\hat{z} - z\|_\infty < \epsilon \text{ iff } |\hat{z}_i - z_i| < \epsilon, \text{ for all } i \in [m].$$

Note that

Lean-expanded library semantic target

```
type:
{Feature : Type u_1} → (Feature → ℝ) → Prop

value:
fun {Feature : Type u_1} (z : Feature → ℝ) => ∀ (i : Feature), |z i| ≤ 1
```

Exact Lean library declaration

```
-- Unit-box convention `z in [-1,1]^m`, stated as absolute-value bounds. -/
def InUnitBox (z : Feature → ℝ) : Prop :=
  ∀ i, |z i| ≤ 1
```

Recorded source-to-library screening Verdict: matches

Reason: The source recovery domain is $z \in [-1, 1]^m$. The Lean predicate requires $|z_i| \leq 1$ for every feature coordinate, which is exactly membership in that unit box.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Math.LinearCompressedSensing.InZeroOne

Source locator: cited publication, lines 700-733

Verbatim paper-source connection

Here, we prove that when feature representations are restricted to the Boolean cube $\{0,1\}^m$, $\Omega(k^2 \log(m/k))$ embedding dimensions are required to linearly separate embeddings with and without a feature. A corollary is that adding a bias and monotonic activation function does not increase the asymptotic capacity of embeddings.

```
\begin{theorem}\label{thm:threshold}
  Suppose that there exist  $A, B \in \mathbb{R}^{d \times m}$  and  $t \in \mathbb{R}^m$  such that for all  $i \in [m]$  and  $k$ -sparse  $z \in \{0,1\}^m$ ,
  \begin{equation}\label{eq:pineapple}
    (B^T A z)_i > t_i
  \end{equation}
  if and only if  $z_i = 1$ . Then for  $k < \sqrt{m}$ ,
  \begin{equation}
    d = \Omega\left(\frac{k^2}{\log k} \log \frac{m}{k}\right).
  \end{equation}
\end{theorem}
It's worth noting here that taking  $\epsilon < \frac{1}{2}$  in \Cref{thm:upper} provides a nearly-matching upper bound of  $\Omega(k^2 \log m)$ , since
 $\epsilon$   $\frac{1}{2}$  implies that  $(B^T A z)_i < \frac{1}{2}$  when  $z_i = 0$  and  $(B^T A z)_i > \frac{1}{2}$  when  $z_i = 1$ . Similarly,
\Cref{thm:threshold} implies \Cref{thm:lower} in the case  $\epsilon < \frac{1}{2}$ . However, the lower bound in \Cref{thm:lower} does not directly
imply \Cref{thm:threshold} since we do not require that  $\|B^T A z - z\|_\infty$  is small here, but rather that  $Az$  linearly separates positive and
negative examples for each feature. That being said, the proof is very similar to that of \Cref{thm:lower}, relying on the same overall strategy.

\begin{proof}
  Suppose that there existed  $A, B \in \mathbb{R}^{d \times m}$  and  $t \in \mathbb{R}^m$  such that for all  $i \in [m]$  and  $k$ -sparse  $z \in \{0,1\}^m$ ,  $(B^T A z)_i > t_i$ 
 $\hookrightarrow$  if and only if  $z_i = 1$ . Then notice that we could scale the  $i$ -th column of  $B$  and  $\gamma_i$  by any constant positive factor to get another
 $\hookrightarrow$  solution. In particular, we can obtain a solution where  $B^T A$  is  $1$  on the diagonal. Therefore, it suffices to show that for  $d \leq$ 
 $\hookrightarrow O(\frac{k^2}{\log k} \log \frac{m}{k})$ , there does not exist  $B, A, t$  satisfying \eqref{eq:pineapple} and such that  $(B^T A)_i = 1$  for all  $i \in$ 
 $\hookrightarrow [m]$ .

  Then we have that  $t_i \leq 1$  since  $(B^T A e)_i > t_i$ . Then, along the lines of the proof of \Cref{thm:lower}, it suffices to find  $i \in [m]$  and
 $\hookrightarrow T \subseteq [m]$  with  $|T| = 2k$  and
  \begin{equation}
    |\angle b_i, a_j| > \frac{1}{k}
  \end{equation}
  for all  $j \in T$ . Indeed, this would imply the existence of a pair  $i \in [m]$ ,  $T \subseteq [m]$  with  $|T| = k$ , such that
  \begin{equation}
    (B^T A z)_i > k \cdot \frac{1}{k} = 1 \geq t_i,
  \end{equation}
  giving a contradiction.

  We will again apply \Cref{lem:alon} to submatrices of  $C$  of size  $r$ . Indeed, \Cref{lem:alon} implies that there exists a constant  $\alpha > 0$  such
 $\hookrightarrow$  that if  $|R| = r$  and
  \begin{equation}
    \text{rank}(C_R) < \alpha \frac{1}{\log r} \left(\frac{1}{k}\right)^2 \log k = \alpha \frac{k^2}{\log k} \log r,
  \end{equation}
  then  $C_R$  has an off-diagonal entry with absolute value at least  $\frac{1}{k}$ . Then we can proceed as in the proof of \Cref{thm:lower}, taking  $r =$ 
 $\hookrightarrow \frac{m}{4k+1}$  and applying Tur'an's theorem.
\end{proof}
```

Lean-expanded library semantic target

```
type:
{Feature : Type u_1} → (Feature → ℝ) → Prop

value:
fun {Feature : Type u_1} (z : Feature → ℝ) => ∀ (i : Feature), z i = 0 ∨ z i = 1
```

Exact Lean library declaration

```
-- Boolean feature vectors, represented over reals. -/
def InZeroOne (z : Feature → ℝ) : Prop :=
  ∀ i, z i = 0 ∨ z i = 1
```

Recorded source-to-library screening Verdict: matches

Reason: The threshold theorem quantifies over Boolean vectors z in $\{0,1\}^m$. The Lean predicate requires every real-valued coordinate to equal either 0 or 1, exactly that source domain.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Math.LinearCompressedSensing.support

Source locator: cited publication, lines 205-209

Verbatim paper-source connection

Let $d(m,k,\epsilon)$ be the smallest choice of d such that there exists $A, B \in \mathbb{R}^{d \times m}$ such that

$$\|B^T A z - z\|_\infty < \epsilon$$

for all k -sparse $z \in [-1, 1]^m$. (Here, any bounded interval would suffice.)

[Next verbatim source excerpt]

Intuition and Proof Techniques

We begin by developing some geometric intuition for the problem. Let $a_1, \dots, a_m \in \mathbb{R}^d$ be the columns of $A \in \mathbb{R}^{d \times m}$ and $b_1, \dots, b_m \in \mathbb{R}^d$ be the columns of $B \in \mathbb{R}^{d \times m}$. For each $i \in [m]$, we recall that a_i is feature i 's **representation vector** and b_i is feature i 's **probe vector**. Now let $\hat{z} = B^T A z$. We are interested in when

$$\|\hat{z} - z\|_\infty < \epsilon \text{ iff } |\hat{z}_i - z_i| < \epsilon, \text{ for all } i \in [m].$$

Note that

Lean-expanded library semantic target

```
type:
{Feature : Type u_1} → [Fintype Feature] → [DecidableEq Feature] → (Feature → ℝ) → Finset Feature

value:
fun {Feature : Type u_1} [Fintype Feature] [DecidableEq Feature] (z : Feature → ℝ) => {i : Feature | z i ≠ 0}
```

Exact Lean library declaration

```
-- Nonzero support of a finite real vector. -/
noncomputable def support [Fintype Feature] [DecidableEq Feature]
  (z : Feature → ℝ) : Finset Feature :=
  Finset.univ.filter fun i => z i ≠ 0
```

Recorded source-to-library screening Verdict: matches

Reason: The source notion of k -sparsity counts nonzero feature coordinates. The Lean support is precisely the finite set of indices i with $z_i \neq 0$ on the paper's finite feature set.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Math.LinearCompressedSensing.KSparse

Source locator: cited publication, lines 205-209

Verbatim paper-source connection

Let $d(m,k,\epsilon)$ be the smallest choice of d such that there exists $A, B \in \mathbb{R}^{d \times m}$ such that

$$\|B^T A z - z\|_\infty < \epsilon$$

for all k -sparse $z \in [-1, 1]^m$. (Here, any bounded interval would suffice.)

[Next verbatim source excerpt]

Intuition and Proof Techniques

We begin by developing some geometric intuition for the problem. Let $a_1, \dots, a_m \in \mathbb{R}^d$ be the columns of $A \in \mathbb{R}^{d \times m}$ and $b_1, \dots, b_m \in \mathbb{R}^d$ be the columns of $B \in \mathbb{R}^{d \times m}$. For each $i \in [m]$, we recall that a_i is feature i 's **representation vector** and b_i is feature i 's **probe vector**. Now let $\hat{z} = B^T A z$. We are interested in when

$$\|\hat{z} - z\|_\infty < \epsilon \text{ iff } |\hat{z}_i - z_i| < \epsilon, \text{ for all } i \in [m].$$

Note that

Lean-expanded library semantic target

```
type:
{Feature : Type u_1} → [Fintype Feature] → [DecidableEq Feature] → ℕ → (Feature → ℝ) → Prop

value:
fun {Feature : Type u_1} [Fintype Feature] [DecidableEq Feature] (k : ℕ) (z : Feature → ℝ) =>
  (AppliedModelingLib.Math.LinearCompressedSensing.support z).card ≤ k
```

Exact Lean library declaration

```
/-- A finite vector is `k`-sparse when its nonzero support has cardinality at most `k`. -/
def KSparse [Fintype Feature] [DecidableEq Feature]
  (k : ℕ) (z : Feature → ℝ) : Prop :=
  (support z).card ≤ k
```

Recorded source-to-library screening Verdict: matches

Reason: The source uses k -sparse vectors, meaning at most k nonzero coordinates. The Lean predicate requires the displayed nonzero support `Finset` to have cardinality at most k ; its finite and decidable-equality binders are available on the paper's $[m]$ index set.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Source locator: cited publication, lines 638-655

Verbatim paper-source connection

```
\begin{proposition}\label{prop:geometry-2}
Consider any constants  $\epsilon > 0$ ,  $\gamma \geq 1$ . Then if  $\|a_i\|, \|b_i\| \leq \gamma$  for all  $i \in [m]$  and
\begin{equation}
\|B^T A z - z\|_\infty < \epsilon
\end{equation}
for all  $k$ -sparse  $z \in [-1, 1]^m$ , it holds that
\begin{enumerate}
\item[(i)]  $\angle \frac{a_i}{\|a_i\|}, \frac{b_i}{\|b_i\|} \geq \frac{1-\epsilon}{\gamma^2}$  for all  $i \in [m]$ ,
\item[(ii)]  $\angle \frac{a_i}{\|a_i\|}, \frac{a_j}{\|a_j\|} \leq \frac{\epsilon \gamma^2}{1-\epsilon} + \sqrt{1 - \frac{(1-\epsilon)^2}{\gamma^4}}$  for all  $i \neq j$ ,
\item[(iii)]  $\angle \frac{b_i}{\|b_i\|}, \frac{b_j}{\|b_j\|} \leq \frac{\epsilon \gamma^2}{1-\epsilon} + \sqrt{1 - \frac{(1-\epsilon)^2}{\gamma^4}}$  for all  $i \neq j$ .
\end{enumerate}
\end{proposition}

\begin{proof}
We must have that  $\angle a_i, b_i \geq 1 - \epsilon$  since  $\|B^T A e_i - e_i\|_\infty \leq \epsilon$ . It follows directly that
\begin{equation}
\angle a_i, b_i > 1 - \epsilon \implies \left| \angle \frac{a_i}{\|a_i\|}, \frac{b_i}{\|b_i\|} \right| \leq \frac{1-\epsilon}{\gamma^2},
\end{equation}
```

Settled source reading Geometry Proposition 2: $\epsilon > 0$ becomes $0 < \epsilon < 1$; the first denominator in parts (ii)/(iii) changes from $1 - \epsilon$ to $(1 - \epsilon)^2$, giving a weaker finite bound.

Lean-expanded library semantic target

```
type:
{Coord : Type u_2} → [Fintype Coord] → (Coord → ℝ) → (Coord → ℝ) → ℝ

value:
fun {Coord : Type u_2} [Fintype Coord] (x y : Coord → ℝ) => ∑ r : Coord, x r * y r
```

Exact Lean library declaration

```
/-- Finite real inner product written as a sum over coordinates. -/
def inner [Fintype Coord] (x y : Coord → ℝ) : ℝ :=
  ∑ r, x r * y r
```

Recorded source-to-library screening Verdict: matches

Reason: Proposition 2 uses the ordinary real Euclidean inner product of d-dimensional columns. The Lean declaration is the finite coordinate sum of $x_r * y_r$, exactly that inner product.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Math.LinearCompressedSensing.linearProbe

Source locator: cited publication, lines 205-209

Verbatim paper-source connection

Let $d(m,k,\epsilon)$ be the smallest choice of d such that there exists $A, B \in \mathbb{R}^{d \times m}$ such that

$$\|B^T A z - z\|_\infty < \epsilon$$

for all k -sparse $z \in [-1, 1]^m$. (Here, any bounded interval would suffice.)

[Next verbatim source excerpt]

Intuition and Proof Techniques

We begin by developing some geometric intuition for the problem. Let $a_1, \dots, a_m \in \mathbb{R}^d$ be the columns of $A \in \mathbb{R}^{d \times m}$ and $b_1, \dots, b_m \in \mathbb{R}^d$ be the columns of $B \in \mathbb{R}^{d \times m}$. For each $i \in [m]$, we recall that a_i is feature i 's **representation vector** and b_i is feature i 's **probe vector**. Now let $\hat{z} = B^T A z$. We are interested in when

$$\|\hat{z} - z\|_\infty < \epsilon \iff |\hat{z}_i - z_i| < \epsilon, \quad \forall i \in [m].$$

Note that

Lean-expanded library semantic target

```
type:
{Feature : Type u_1} →
{Coord : Type u_2} →
[Fintype Feature] → [Fintype Coord] → (Feature → Coord → ℝ) → (Feature → Coord → ℝ) → (Feature → ℝ) → Feature → ℝ

value:
fun {Feature : Type u_1} {Coord : Type u_2} [Fintype Feature] [Fintype Coord] (A B : Feature → Coord → ℝ)
(z : Feature → ℝ) (i : Feature) =>
Σ j : Feature, z j * AppliedModelingLib.Math.LinearCompressedSensing.inner (B i) (A j)
```

Exact Lean library declaration

```
/-- Coordinate `i` of `B^T A z`, written using representation and probe columns. -/
def linearProbe [Fintype Feature] [Fintype Coord]
(A B : Feature → Coord → ℝ) (z : Feature → ℝ) (i : Feature) : ℝ :=
Σ j, z j * inner (B i) (A j)
```

Recorded source-to-library screening **Verdict:** matches

Reason: The source defines $\hat{z} = B^T A z$. Its i -th coordinate is $\sum_j z_j$ times the inner product of probe column B_i with representation column A_j , which is exactly the displayed Lean formula.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Math.LinearCompressedSensing.SuperErrorLt

Source locator: cited publication, lines 205-209

Verbatim paper-source connection

Let $d(m,k,\epsilon)$ be the smallest choice of d such that there exists $A, B \in \mathbb{R}^{d \times m}$ such that

$$\|B^T A z - z\|_\infty < \epsilon$$

for all k -sparse $z \in [-1, 1]^m$. (Here, any bounded interval would suffice.)

[Next verbatim source excerpt]

Intuition and Proof Techniques

We begin by developing some geometric intuition for the problem. Let $a_1, \dots, a_m \in \mathbb{R}^d$ be the columns of $A \in \mathbb{R}^{d \times m}$ and $b_1, \dots, b_m \in \mathbb{R}^d$ be the columns of $B \in \mathbb{R}^{d \times m}$. For each $i \in [m]$, we recall that a_i is feature i 's **representation vector** and b_i is feature i 's **probe vector**. Now let $\hat{z} = B^T A z$. We are interested in when

$$\|\hat{z} - z\|_\infty < \epsilon \iff |\hat{z}_i - z_i| < \epsilon, \text{ for all } i \in [m].$$

Note that

Lean-expanded library semantic target

```
type:
{Feature : Type u_1} →
{Coord : Type u_2} →
[Fintype Feature] → [Fintype Coord] → (Feature → Coord → ℝ) → (Feature → Coord → ℝ) → (Feature → ℝ) → ℝ → Prop

value:
fun {Feature : Type u_1} {Coord : Type u_2} [Fintype Feature] [Fintype Coord] (A B : Feature → Coord → ℝ)
(z : Feature → ℝ) (ε : ℝ) =>
  ∀ (i : Feature), |linearProbe A B z i - z i| < ε
```

Exact Lean library declaration

```
-- Paper-style coordinatewise  $\ell_\infty$  recovery error. -/
def SuperErrorLt [Fintype Feature] [Fintype Coord]
  (A B : Feature → Coord → ℝ) (z : Feature → ℝ) (ε : ℝ) : Prop :=
  ∀ i, |linearProbe A B z i - z i| < ε
```

Recorded source-to-library screening Verdict: matches

Reason: The source explicitly equates $\|B^T A z - z\|_\infty < \epsilon$ with every coordinate error being strictly below ϵ . The Lean predicate states for all i , $|\text{linearProbe } A \ B \ z \ i - z \ i| < \epsilon$, with the same strict bound and quantification.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Math.LinearCompressedSensing.ThresholdSeparates

Source locator: cited publication, lines 700-733

Verbatim paper-source connection

Here, we prove that when feature representations are restricted to the Boolean cube $\{0,1\}^m$, $\Omega(k^2 \log(m/k))$ embedding dimensions are required to linearly separate embeddings with and without a feature. A corollary is that adding a bias and monotonic activation function does not increase the asymptotic capacity of embeddings.

```
\begin{theorem}\label{thm:threshold}
  Suppose that there exist  $A, B \in \mathbb{R}^{d \times m}$  and  $t \in \mathbb{R}^m$  such that for all  $i \in [m]$  and  $k$ -sparse  $z \in \{0,1\}^m$ ,
  \begin{equation}\label{eq:pineapple}
    (B^T A z)_i > t_i
  \end{equation}
  if and only if  $z_i = 1$ . Then for  $k < \sqrt{m}$ ,
  \begin{equation}
    d = \Omega\left(\frac{k^2}{\log k} \log \frac{m}{k}\right).
  \end{equation}
\end{theorem}
It's worth noting here that taking  $\epsilon < \frac{1}{2}$  in \Cref{thm:upper} provides a nearly-matching upper bound of  $\Omega(k^2 \log m)$ , since
 $\epsilon < \frac{1}{2}$  implies that  $(B^T A z)_i < \frac{1}{2}$  when  $z_i = 0$  and  $(B^T A z)_i > \frac{1}{2}$  when  $z_i = 1$ . Similarly,
\Cref{thm:threshold} implies \Cref{thm:lower} in the case  $\epsilon < \frac{1}{2}$ . However, the lower bound in \Cref{thm:lower} does not directly
imply \Cref{thm:threshold} since we do not require that  $\|B^T A z - z\|_\infty$  is small here, but rather that  $Az$  linearly separates positive and
negative examples for each feature. That being said, the proof is very similar to that of \Cref{thm:lower}, relying on the same overall strategy.

\begin{proof}
  Suppose that there existed  $A, B \in \mathbb{R}^{d \times m}$  and  $t \in \mathbb{R}^m$  such that for all  $i \in [m]$  and  $k$ -sparse  $z \in \{0,1\}^m$ ,  $(B^T A z)_i > t_i$ 
  if and only if  $z_i = 1$ . Then notice that we could scale the  $i$ -th column of  $B$  and  $\gamma_i$  by any constant positive factor to get another
  solution. In particular, we can obtain a solution where  $B^T A$  is  $1$  on the diagonal. Therefore, it suffices to show that for  $d \leq
  O(\frac{k^2}{\log k} \log \frac{m}{k})$ , there does not exist  $B, A, t$  satisfying \eqref{eq:pineapple} and such that  $(B^T A)_{ii} = 1$  for all  $i \in
  [m]$ .

  Then we have that  $t_i \leq 1$  since  $(B^T A e_i)_i > t_i$ . Then, along the lines of the proof of \Cref{thm:lower}, it suffices to find  $i \in [m]$  and
   $T^* \subseteq [m]$  with  $|T^*| = 2k$  and
  \begin{equation}
    |\angle b_i, a_j| > \frac{1}{k}
  \end{equation}
  for all  $j \in T^*$ . Indeed, this would imply the existence of a pair  $i \in [m]$ ,  $T \subseteq [m]$  with  $|T| = k$ , such that
  \begin{equation}
    (B^T A z)_i > k \cdot \frac{1}{k} = 1 \geq t_i,
  \end{equation}
  giving a contradiction.

  We will again apply \Cref{lem:alon} to submatrices of  $C$  of size  $r$ . Indeed, \Cref{lem:alon} implies that there exists a constant  $\alpha > 0$  such
  that if  $|R| = r$  and
  \begin{equation}
    \text{rank}(C_R) < \alpha \frac{1}{\log r} \left(\frac{1}{k}\right)^2 \log k = \alpha \frac{k^2}{\log k} \log r,
  \end{equation}
  then  $C_R$  has an off-diagonal entry with absolute value at least  $\frac{1}{k}$ . Then we can proceed as in the proof of \Cref{thm:lower}, taking  $r =
  \frac{1}{\log k}$  and applying Tur'an's theorem.
\end{proof}
```

Lean-expanded library semantic target

```
type:
{Feature : Type u_1} →
{Coord : Type u_2} →
[Fintype Feature] →
[DecidableEq Feature] → [Fintype Coord] → (Feature → Coord → ℝ) → (Feature → Coord → ℝ) → (Feature → ℝ) → ℕ → Prop

value:
fun {Feature : Type u_1} {Coord : Type u_2} [Fintype Feature] [DecidableEq Feature] [Fintype Coord]
(A B : Feature → Coord → ℝ) (threshold : Feature → ℝ) (k : ℕ) =>
∀ (i : Feature) (z : Feature → ℝ),
  AppliedModelingLib.Math.LinearCompressedSensing.KSparse k z →
  AppliedModelingLib.Math.LinearCompressedSensing.InZeroOne z →
  (threshold i < linearProbe A B z i ↔ z i = 1)
```

Exact Lean library declaration

```
/-
Threshold separation for Boolean feature vectors: row `i` crosses threshold
exactly on `k`-sparse Boolean vectors with feature `i` present.
-/
def ThresholdSeparates [Fintype Feature] [DecidableEq Feature] [Fintype Coord]
(A B : Feature → Coord → ℝ) (threshold : Feature → ℝ) (k : ℕ) : Prop :=
∀ i (z : Feature → ℝ), KSparse k z → InZeroOne z →
(threshold i < linearProbe A B z i ↔ z i = 1)
```

Recorded source-to-library screening Verdict: matches

Reason: The source requires $(B^T A z)_i > t_i$ iff $z_i = 1$ for every feature and k -sparse Boolean z . The Lean predicate has the same universal binders, $KSparse$ and $InZeroOne$ premises, and the equivalent strict inequality $\text{threshold}_i < \text{linearProbe}_i$ iff $z_i = 1$.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Math.LinearCompressedSensing.vectorNorm

Source locator: cited publication, lines 638-655

Verbatim paper-source connection

```
\begin{proposition}\label{prop:geometry-2}
Consider any constants  $\epsilon > 0$ ,  $\gamma \geq 1$ . Then if  $\|\mathbf{a}_i\|, \|\mathbf{b}_i\| \leq \gamma$  for all  $i \in [m]$  and
\begin{equation}
\|\mathbf{B}^T \mathbf{A} \mathbf{z} - \mathbf{z}\|_\infty < \epsilon
\end{equation}
for all  $\mathbf{z}$   $k$ -sparse  $\mathbf{z} \in [-1, 1]^m$ , it holds that
\begin{enumerate}
\item[(i)]  $\angle \frac{\mathbf{a}_i}{\|\mathbf{a}_i\|}, \frac{\mathbf{b}_i}{\|\mathbf{b}_i\|} \geq \frac{1-\epsilon}{\gamma^2}$  for all  $i \in [m]$ ,
\item[(ii)]  $\angle \frac{\mathbf{a}_i}{\|\mathbf{a}_i\|}, \frac{\mathbf{a}_j}{\|\mathbf{a}_j\|} \leq \frac{\epsilon \gamma^2}{1-\epsilon} + \sqrt{1 - \frac{(1-\epsilon)^2}{\gamma^4}}$  for all  $i \neq j$ ,
\item[(iii)]  $\angle \frac{\mathbf{b}_i}{\|\mathbf{b}_i\|}, \frac{\mathbf{b}_j}{\|\mathbf{b}_j\|} \leq \frac{\epsilon}{\gamma^2} + \sqrt{1 - \frac{(1-\epsilon)^2}{\gamma^4}}$  for all  $i \neq j$ .
\end{enumerate}
\end{proposition}

\begin{proof}
We must have that  $\angle \mathbf{a}_i, \mathbf{b}_i \geq 1 - \epsilon$  since  $\|\mathbf{B}^T \mathbf{A} \mathbf{e}_i - \mathbf{e}_i\|_\infty \leq \epsilon$ . It follows directly that
\begin{equation}
\angle \mathbf{a}_i, \mathbf{b}_i > 1 - \epsilon \implies \left| \angle \frac{\mathbf{a}_i}{\|\mathbf{a}_i\|}, \frac{\mathbf{b}_i}{\|\mathbf{b}_i\|} \right| \leq \frac{1-\epsilon}{\gamma^2},
\end{equation}
```

Settled source reading Geometry Proposition 2: $\epsilon > 0$ becomes $0 < \epsilon < 1$; the first denominator in parts (ii)/(iii) changes from $1 - \epsilon$ to $(1 - \epsilon)^2$, giving a weaker finite bound.

Lean-expanded library semantic target

```
type:
{Coord : Type u_2} → [Fintype Coord] → (Coord → ℝ) → ℝ

value:
fun {Coord : Type u_2} [Fintype Coord] (x : Coord → ℝ) => AppliedModelingLib.FiniteDimensionalNorms.l2 x
```

Exact Lean library declaration

```
/-- Finite Euclidean column norm associated with `inner`. -/
noncomputable def vectorNorm [Fintype Coord] (x : Coord → ℝ) : ℝ :=
  AppliedModelingLib.FiniteDimensionalNorms.l2 x
```

Recorded source-to-library screening Verdict: matches

Reason: The source column norm is the finite Euclidean norm. The Lean declaration delegates to the displayed `l2` quantity, which is $\sqrt{\sum_r x_r^2}$, so its value agrees exactly on $\mathbb{R}^d$.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Source locator: cited publication, lines 638-680

Verbatim paper-source connection

```
\begin{proposition}\label{prop:geometry-2}
Consider any constants  $\epsilon > 0$ ,  $\gamma \geq 1$ . Then if  $\|a_i\|, \|b_i\| \leq \gamma$  for all  $i \in [m]$  and
\begin{equation}
\|B^T A z - z\|_\infty < \epsilon
\end{equation}
for all  $k$ -sparse  $z \in [-1, 1]^m$ , it holds that
\begin{enumerate}
\item[(i)]  $\left| \left\langle \frac{a_i}{\|a_i\|}, \frac{b_j}{\|b_j\|} \right\rangle - \frac{1-\epsilon}{\gamma^2} \right| \leq \epsilon$  for all  $i \in [m]$ ,
\item[(ii)]  $\left| \left\langle \frac{a_i}{\|a_i\|}, \frac{a_j}{\|a_j\|} \right\rangle - \frac{1-\epsilon}{\gamma^2} \right| \leq \epsilon$  for all  $i \neq j$ ,
\item[(iii)]  $\left| \left\langle \frac{b_i}{\|b_i\|}, \frac{b_j}{\|b_j\|} \right\rangle - \frac{1-\epsilon}{\gamma^2} \right| \leq \epsilon$  for all  $i \neq j$ .
\end{enumerate}
\end{proposition}

\begin{proof}
We must have that  $\left| \left\langle a_i, b_j \right\rangle \right| \geq 1 - \epsilon$  since  $\|B^T A e_i - e_i\|_\infty \leq \epsilon$ . It follows directly that
\begin{equation}
\left| \left\langle a_i, b_j \right\rangle \right| > 1 - \epsilon \implies \left| \left\langle \frac{a_i}{\|a_i\|}, \frac{b_j}{\|b_j\|} \right\rangle - \frac{1-\epsilon}{\gamma^2} \right| \leq \epsilon,
\end{equation}
establishing (i).

We now show (ii). By Cauchy-Schwarz,  $\|a_i\| \|b_j\| \geq \left| \left\langle a_i, b_j \right\rangle \right| > 1 - \epsilon$  for all  $i \in [m]$ . So since
 $\implies \|a_i\|, \|b_j\| \leq \gamma$  for all  $i \in [m]$ , we have that  $\|a_i\|, \|b_j\| \geq \frac{1-\epsilon}{\gamma}$  for all  $i \in [m]$ .
Therefore, we have that
\begin{equation}
\left| \left\langle \frac{a_i}{\|a_i\|}, \frac{a_j}{\|a_j\|} \right\rangle - \frac{1-\epsilon}{\gamma^2} \right| \leq \epsilon
\implies \left| \left\langle \frac{a_i}{\|a_i\|}, \frac{a_j}{\|a_j\|} \right\rangle \right| \leq \frac{1-\epsilon}{\gamma^2} + \epsilon.
\end{equation}
Let  $\tilde{a}_i$  denote  $\frac{a_i}{\|a_i\|}$  and  $\tilde{b}_i$  denote  $\frac{b_i}{\|b_i\|}$  for all  $i \in [m]$ . Then write
\begin{align}
\tilde{a}_i &= \alpha_i \tilde{b}_i + \tilde{a}_i^\perp \\
\tilde{a}_j &= \alpha_j \tilde{b}_j + \tilde{a}_j^\perp,
\end{align}
where  $\alpha_i = \left\langle \tilde{a}_i, \tilde{b}_i \right\rangle$ ,  $\alpha_j = \left\langle \tilde{a}_j, \tilde{b}_j \right\rangle$ , and  $\tilde{a}_i^\perp, \tilde{a}_j^\perp \perp \tilde{b}_i, \tilde{b}_j$ . Then
\begin{equation}
\left| \left\langle \tilde{a}_i, \tilde{a}_j \right\rangle - \frac{1-\epsilon}{\gamma^2} \right| = \left| \alpha_i \alpha_j + \left\langle \tilde{a}_i^\perp, \tilde{a}_j^\perp \right\rangle - \frac{1-\epsilon}{\gamma^2} \right|.
\end{equation}
By Cauchy-Schwarz,
\begin{equation}
\left| \left\langle \tilde{a}_i^\perp, \tilde{a}_j^\perp \right\rangle \right| \leq \|\tilde{a}_i^\perp\| \|\tilde{a}_j^\perp\| = \sqrt{(1-\alpha_i^2)(1-\alpha_j^2)}.
\end{equation}
Therefore,
\begin{equation}
\left| \left\langle \tilde{a}_i, \tilde{a}_j \right\rangle - \frac{1-\epsilon}{\gamma^2} \right| \leq |\alpha_i| |\alpha_j| + \sqrt{(1-\alpha_i^2)(1-\alpha_j^2)}.
\end{equation}
We have that  $|\alpha_i| \geq \frac{1-\epsilon}{\gamma}$  and  $0 < |\alpha_j| < \frac{1-\epsilon}{\gamma}$ . Therefore,
\begin{equation}
\left| \left\langle \tilde{a}_i, \tilde{a}_j \right\rangle - \frac{1-\epsilon}{\gamma^2} \right| \leq \frac{1-\epsilon}{\gamma} + \sqrt{(1-\alpha_i^2)(1-\alpha_j^2)}.
\end{equation}
```

Settled source reading Geometry Proposition 2: $\epsilon > 0$ becomes $0 < \epsilon < 1$; the first denominator in parts (ii)/(iii) changes from $1 - \epsilon$ to $(1 - \epsilon)^2$, giving a weaker finite bound.

Formalized review target The archival source above is not asserted equivalent to this formalized target.

Under the stated recovery and column-norm hypotheses, Geometry Proposition 2(ii) holds with first term $\epsilon * \gamma^2 / (1-\epsilon)^2$ in the normalized representation-vector correlation bound.

Archival source anchor: cited publication, lines 638-680

Lean-expanded library semantic target

```
type:
{Coord : Type u_2} → [Fintype Coord] → (Coord → ℝ) → Coord → ℝ

value:
fun {Coord : Type u_2} [Fintype Coord] (x : Coord → ℝ) (a : Coord) =>
  (AppliedModelingLib.Math.LinearCompressedSensing.vectorNorm x)-1 * x a
```

Exact Lean library declaration

```
-- Normalize a finite vector using `vectorNorm`. Zero vectors normalize to zero. -/
noncomputable def normalizedVector [Fintype Coord] (x : Coord → ℝ) : Coord → ℝ :=
  fun r => (vectorNorm x)-1 * x r
```

Recorded source-to-library screening Verdict: matches_approved_corrected_target

Reason: The selected item is explicitly bound to the documented, materially non-equivalent correction LRH-GEOMETRY2-DENOMINATOR-01: under the stated recovery and column-norm hypotheses, Geometry Proposition 2(ii)'s normalized representation-column correlation bound has first term $\epsilon * \gamma^2 / (1-\epsilon)^2$, rather than the archival $\epsilon * \gamma^2 / (1-\epsilon)$. Compared with that corrected target under the displayed approved $0 < \epsilon < 1$ reading, the expanded target sends each coordinate of finite x to $(\sqrt{\sum_r x_r^2})^{-1} * x_r$, exactly the source normalization $a_i / \text{norm}(a_i)$ appearing on the corrected bound's left side. The approved context supplies positive column norms, and the displayed paper use restricts this generic operation to finite feature-matrix columns, so its zero-vector value adds no source-domain case. The corrected denominator is a bound term rather than an alternative normalization formula; this prerequisite preserves the corrected target's normalized representation-column scope without adding a premise or changing a formula.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Source locator: cited publication, lines 638-680

Verbatim paper-source connection

```
\begin{proposition}\label{prop:geometry-2}
Consider any constants  $\epsilon > 0$ ,  $\gamma \geq 1$ . Then if  $\|a_i\|, \|b_i\| \leq \gamma$  for all  $i \in [m]$  and
\begin{equation}
\|B^T A z - z\|_\infty < \epsilon
\end{equation}
for all  $k$ -sparse  $z \in [-1, 1]^m$ , it holds that
\begin{enumerate}
\item(i)  $\angle \frac{a_i}{\|a_i\|}, \frac{b_i}{\|b_i\|} \leq \frac{1-\epsilon}{\gamma^2}$  for all  $i \in [m]$ ,
\item(ii)  $\angle \frac{a_i}{\|a_i\|}, \frac{a_j}{\|a_j\|} \leq \frac{\epsilon \gamma^2}{1-\epsilon} + \sqrt{1 - \frac{\epsilon}{\gamma^2}}$  for all  $i \neq j$ ,
\item(iii)  $\angle \frac{b_i}{\|b_i\|}, \frac{b_j}{\|b_j\|} \leq \frac{\epsilon \gamma^2}{1-\epsilon} + \sqrt{1 - \frac{\epsilon}{\gamma^2}}$  for all  $i \neq j$ .
\end{enumerate}
\end{proposition}

\begin{proof}
We must have that  $\angle a_i, b_i \leq 1 - \epsilon$  since  $\|B^T A e_i - e_i\|_\infty \leq \epsilon$ . It follows directly that
\begin{equation}
\angle a_i, b_i > 1 - \epsilon \implies \left\| \frac{a_i}{\|a_i\|}, \frac{b_i}{\|b_i\|} \right\| \leq \frac{1-\epsilon}{\gamma^2},
\end{equation}
establishing (i).

We now show (ii). By Cauchy-Schwarz,  $\|a_i\| \|b_i\| \geq \angle a_i, b_i > 1 - \epsilon$  for all  $i \in [m]$ . So since
 $\|a_i\|, \|b_i\| \leq \gamma$  for all  $i \in [m]$ , we have that  $\|a_i\|, \|b_i\| \geq \frac{1-\epsilon}{\gamma}$  for all  $i \in [m]$ .
Therefore, we have that
\begin{equation}
\left\| \frac{b_i}{\|b_i\|}, \frac{a_j}{\|a_j\|} \right\| \leq \left\| \frac{b_i}{\|b_i\|}, a_j \right\| \frac{\gamma}{1-\epsilon} <
\frac{\epsilon \gamma^2}{1-\epsilon}.
\end{equation}
Let  $\tilde{a}_i$  denote  $\frac{a_i}{\|a_i\|}$  and  $\tilde{b}_i$  denote  $\frac{b_i}{\|b_i\|}$  for all  $i \in [m]$ . Then write
\begin{align}
\tilde{a}_i &= \alpha_i \tilde{b}_i + \tilde{a}_i^\perp \\
\tilde{a}_j &= \alpha_j \tilde{b}_i + \tilde{a}_j^\perp,
\end{align}
where  $\alpha_i = \angle \tilde{a}_i, \tilde{b}_i$ ,  $\alpha_j = \angle \tilde{a}_j, \tilde{b}_i$ , and  $\tilde{a}_i^\perp, \tilde{a}_j^\perp$ 
 $\perp \tilde{b}_i$ . Then
\begin{equation}
\angle \tilde{a}_i, \tilde{a}_j = \alpha_i \alpha_j + \angle \tilde{a}_i^\perp, \tilde{a}_j^\perp.
\end{equation}
By Cauchy-Schwarz,
\begin{equation}
|\angle \tilde{a}_i^\perp, \tilde{a}_j^\perp| \leq \|\tilde{a}_i^\perp\| \|\tilde{a}_j^\perp\| = \sqrt{(1-\alpha_i^2)(1-\alpha_j^2)}.
\end{equation}
Therefore,
\begin{equation}
|\angle \tilde{a}_i, \tilde{a}_j| \leq |\alpha_i \alpha_j| + \sqrt{(1-\alpha_i^2)(1-\alpha_j^2)}.
\end{equation}
We have that  $\alpha_i > \frac{1-\epsilon}{\gamma^2}$  and  $0 < \alpha_j < \frac{\epsilon \gamma^2}{1-\epsilon}$ . Therefore,
```

Settled source reading Geometry Proposition 2: $\epsilon > 0$ becomes $0 < \epsilon < 1$; the first denominator in parts (ii)/(iii) changes from $1 - \epsilon$ to $(1 - \epsilon)^2$, giving a weaker finite bound.

Formalized review target The archival source above is not asserted equivalent to this formalized target.

Under the stated recovery and column-norm hypotheses, Geometry Proposition 2(iii) holds with first term $\epsilon * \gamma^2 / (1-\epsilon)^2$ in the normalized probe-vector correlation bound.

Archival source anchor: cited publication, lines 638-680

Lean-expanded library semantic target

```
type:
{Coord : Type u_2} → [Fintype Coord] → (Coord → ℝ) → (Coord → ℝ) → ℝ

value:
fun {Coord : Type u_2} [Fintype Coord] (x y : Coord → ℝ) =>
  AppliedModelingLib.Math.LinearCompressedSensing.inner
    (AppliedModelingLib.Math.LinearCompressedSensing.normalizedVector x)
    (AppliedModelingLib.Math.LinearCompressedSensing.normalizedVector y)
```

Exact Lean library declaration

```
/-- Finite Euclidean correlation of two vectors. -/
noncomputable def vectorCorrelation [Fintype Coord]
  (x y : Coord → ℝ) : ℝ :=
  inner (normalizedVector x) (normalizedVector y)
```

Recorded source-to-library screening Verdict: matches_approved_corrected_target

Reason: The selected item is explicitly bound to the documented, materially non-equivalent correction LRH-GEOMETRY2-DENOMINATOR-01: under the stated recovery and column-norm hypotheses, Geometry Proposition 2(iii)'s normalized probe-vector correlation bound has first term $\epsilon * \gamma^2 / (1-\epsilon)^2$, rather than the archival $\epsilon * \gamma^2 / (1-\epsilon)$. Compared with that corrected target under the displayed approved $0 < \epsilon < 1$ reading, the expanded target is $\sum_r ((\sqrt{\sum_s x_{s^2}})^{-1} * x_r) * ((\sqrt{\sum_s y_{s^2}})^{-1} * y_r)$, exactly the finite inner product of the normalized probe columns $\bar{b}_i / \text{norm}(\bar{b}_i)$ and $\bar{b}_j / \text{norm}(\bar{b}_j)$ on the corrected bound's left side. The displayed paper use restricts the generic finite-coordinate operation to feature-matrix columns, and the approved context supplies the positive norms required there. Thus this definition preserves the approved corrected target's normalized probe-correlation formula and domain without adding a premise or changing either its left side or its corrected bound scope.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Math.TendsToZero

Source locator: cited publication, lines 601-632

Verbatim paper-source connection

```

\begin{align}
& \left| \left| \angle \frac{a_i}{\|a_i\|}, \frac{b_i}{\|b_i\|} \right| \right| \\
& = \frac{\left| \angle c_i + \lambda a^*, c_i + \lambda b^* \right|}{\|a_i\| \|b_i\|} \\
& \leq \frac{\left| \angle c_i, c_i \right| + \left| \angle c_i, \lambda b^* \right| + \left| \angle \lambda a^*, c_i \right| + \left| \angle \lambda a^*, \lambda b^* \right|}{\|a_i\| \|b_i\|} \\
& \leq \frac{1 + (2\lambda + \lambda^2)\mu}{\|a_i\| \|b_i\|} = \frac{1 + \frac{2\lambda}{1 + \lambda^2}}{\|a_i\| \|b_i\|} \text{.} \text{\label{eq:dog}}
\end{align}

```

Note that

```

\begin{align}
& \angle a_i, a_i \angle = \angle c_i + \lambda a^*, c_i + \lambda a^* \angle \\
& \leq \angle a^*, \lambda a^* \angle \\
& \leq 1 + \lambda^2 - 2\lambda\mu = 1 + \lambda^2 - \frac{2\lambda}{1 + \lambda^2} = 1 + \lambda^2 - o(1).
\end{align}

```

Therefore, $\|a_i\| \leq \sqrt{1 + \lambda^2 - o(1)}$. Analogously, $\|b_i\| \leq \sqrt{1 + \lambda^2 - o(1)}$.
So we have, plugging into \eqref{eq:dog},

```

\begin{align}
& \left| \angle \frac{a_i}{\|a_i\|}, \frac{b_i}{\|b_i\|} \right| \\
& \leq \frac{1 + o(1)}{1 + \lambda^2 - o(1)} = \frac{1}{1 + \lambda^2} + o(1) = \delta + o(1),
\end{align}

```

where we recall that we set $\lambda = \sqrt{\frac{1}{\delta} - 1}$.

\paragraph{Establishing (ii,iii).} We now show that the construction satisfies (ii). (iii) follows analogously. For all $i \neq j$,

```

\begin{align}
& \left| \left| \angle \frac{a_i}{\|a_i\|}, \frac{a_j}{\|a_j\|} \right| \right| \\
& = \frac{\left| \angle c_i + \lambda a^*, c_j + \lambda a^* \right|}{\|a_i\| \|a_j\|} \\
& \leq \frac{\left| \angle c_i, c_j \right| + \left| \angle c_i, \lambda a^* \right| + \left| \angle \lambda a^*, c_j \right| + \left| \angle \lambda a^*, \lambda a^* \right|}{\|a_i\| \|a_j\|} \\
& \leq \frac{\lambda^2 - (1 + 2\lambda)\mu}{\|a_i\| \|a_j\|} = \frac{1 + \lambda^2 + o(1)}{\|a_i\| \|a_j\|} \\
& = \frac{\lambda^2 - o(1)}{1 + \lambda^2 + o(1)} = 1 - \frac{1}{1 + \lambda^2} - o(1) = 1 - \delta - o(1),
\end{align}

```

where we again recall that where we recall that $\lambda = \sqrt{\frac{1}{\delta} - 1}$.

Lean-expanded library semantic target

```

type:
  (ℕ → ℝ) → Prop

value:
  fun (ε : ℕ → ℝ) => Filter.Tendsto ε Filter.atTop (nhds 0)

```

Exact Lean library declaration

```

/-- A sequence tends to zero. -/
def TendsToZero (ε : ℕ → ℝ) : Prop :=
  Tendsto ε atTop (nhds 0)

```

Recorded source-to-library screening Verdict: matches

Reason: The Geometry Proposition 1 source uses $o(1)$ errors as k grows. Lean states convergence of a real sequence to 0 along the natural-number `atTop` filter, which is exactly the required asymptotic meaning.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Probability.RademacherMatrix.rademacherSign

Source locator: cited publication, lines 412-418

Verbatim paper-source connection

```
\begin{lemma}[Incoherence of Random Matrices]\label{prop:incoherent}
  Let  $C \in \mathbb{R}^{d \times m}$  be a matrix with i.i.d. scaled Rademacher entries  $\pm \frac{1}{\sqrt{d}}$ . For
  \begin{equation}
    d = O\left(\frac{\log m + \log \frac{1}{\delta}}{\mu^2}\right),
  \end{equation}
   $C$  is  $\mu$ -incoherent with probability at least  $1 - \delta$ .
\end{lemma}
```

Lean-expanded library semantic target

```
type:
  Bool → ℝ

value:
  fun (b : Bool) => if b = true then 1 else -1
```

Exact Lean library declaration

```
/-- The symmetric Rademacher sign encoded by a Boolean. -/
def rademacherSign (b : Bool) : ℝ :=
  if b then 1 else -1
```

Recorded source-to-library screening Verdict: matches

Reason: The random-matrix source uses Rademacher signs taking values +1 and -1. The Lean declaration maps one Boolean outcome to 1 and the other to -1, exactly those two signs.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Probability.RademacherMatrix.rowsMeasure

Source locator: cited publication, lines 412-418

Verbatim paper-source connection

```
\begin{lemma}[Incoherence of Random Matrices]\label{prop:incoherent}
  Let  $C \in \mathbb{R}^{d \times m}$  be a matrix with i.i.d. scaled Rademacher entries  $\pm \frac{1}{\sqrt{d}}$ . For
  \begin{equation}
    d = O\left(\frac{\log m + \log \frac{1}{\delta}}{\mu^2}\right),
  \end{equation}
   $C$  is  $\mu$ -incoherent with probability at least  $1 - \delta$ .
\end{lemma}
```

Lean-expanded library semantic target

```
type:
  (Feature : Type u_3) → (Coord : Type u_4) → MeasureTheory.Measure (Coord → Feature → Bool)

value:
  fun (Feature : Type u_3) (Coord : Type u_4) =>
    MeasureTheory.Measure.infinitePi fun (x : Coord) => AppliedModelingLib.FairCoin.productMeasure Feature
```

Exact Lean library declaration

```
/--
Row-product fair-coin law for a random `Coord × Feature` Boolean matrix.
Rows are sampled independently, and each row is a fair Boolean vector over
features.
-/
noncomputable def rowsMeasure (Feature Coord : Type*) :
  Measure (Coord → Feature → Bool) :=
  Measure.infinitePi fun _ : Coord => FairCoin.productMeasure Feature
```

Recorded source-to-library screening Verdict: matches

Reason: The source requires i.i.d. Rademacher entries. The displayed Lean law is an independent product over coordinates of fair-coin product laws over features, so on $\text{Fin } d$ by $\text{Fin } m$ it gives independent fair Boolean entries as required.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.Probability.RademacherMatrix.scaledMatrix

Source locator: cited publication, lines 412-418

Verbatim paper-source connection

```
\begin{lemma}[Incoherence of Random Matrices]\label{prop:incoherent}
  Let  $C \in \mathbb{R}^{d \times m}$  be a matrix with i.i.d. scaled Rademacher entries  $\pm \frac{1}{\sqrt{d}}$ . For
  \begin{equation}
    d = O\left(\frac{\log m + \log \frac{1}{\delta}}{\mu^2}\right),
  \end{equation}
   $C$  is  $\mu$ -incoherent with probability at least  $1 - \delta$ .
\end{lemma}
```

Lean-expanded library semantic target

```
type:
{Feature : Type u_1} → {Coord : Type u_2} → [Fintype Coord] → (Coord → Feature → Bool) → Feature → Coord → ℝ

value:
fun {Feature : Type u_1} {Coord : Type u_2} [Fintype Coord] (ω : Coord → Feature → Bool) (a : Feature) (a_1 : Coord) =>
  (√ ↑ (Fintype.card Coord))-1 * AppliedModelingLib.Probability.RademacherMatrix.rademacherSign (ω a_1 a)
```

Exact Lean library declaration

```
/-- A scaled Rademacher matrix, with columns indexed by `Feature`. -/
noncomputable def scaledMatrix [Fintype Coord]
  (ω : Coord → Feature → Bool) : Feature → Coord → ℝ :=
  fun j r =>
    (Real.sqrt (Fintype.card Coord : ℝ))-1 * rademacherSign (ω r j)
```

Recorded source-to-library screening Verdict: matches

Reason: The source matrix has entries plus or minus $1/\sqrt{d}$. Lean multiplies each displayed Rademacher sign by the inverse square root of the coordinate-cardinality d ; the approved construction has $d > 0$, so this is exactly the stated scaling.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

AppliedModelingLib.measureProb

Source locator: cited publication, lines 412-418

Verbatim paper-source connection

```
\begin{lemma}[Incoherence of Random Matrices]\label{prop:incoherent}
  Let  $C \in \mathbb{R}^{d \times m}$  be a matrix with i.i.d. scaled Rademacher entries  $\frac{1}{\sqrt{d}}$ . For
  \begin{equation}
    d = O\left(\frac{\log m + \log \frac{1}{\delta}}{\mu^2}\right),
  \end{equation}
   $C$  is  $\mu$ -incoherent with probability at least  $1 - \delta$ .
\end{lemma}
```

Lean-expanded library semantic target

```
type:
{α : Type u_1} → [inst : MeasurableSpace α] → MeasureTheory.Measure α → (α → Prop) → ℝ

value:
fun {α : Type u_1} [MeasurableSpace α] (μ : MeasureTheory.Measure α) (p : α → Prop) => (μ {a : α | p a}).toReal
```

Exact Lean library declaration

```
/-- Real-valued probability/mass of an event under a measure. -/
noncomputable def measureProb {α : Type*} [MeasurableSpace α]
  (μ : Measure α) (p : α → Prop) : ℝ :=
  (μ {a | p a}).toReal
```

Recorded source-to-library screening Verdict: matches

Reason: The source conclusion compares the probability of the random-matrix event with $1 - \delta$. Lean takes the real value of the event's measure; on the displayed finite Rademacher probability measure this is exactly the real-valued event probability used by that conclusion.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Paper-specific semantic prerequisites

GKP26LinearRepresentationHypothesis.linearCompressedSensingDimensionValue

Paper-local declaration:

GKP26LinearRepresentationHypothesis.linearCompressedSensingDimensionValue

Source locator: cited publication, lines 205-209

Verbatim paper-source connection

Let $d(m,k,\epsilon)$ be the smallest choice of d such that there exists $A,B \in \mathbb{R}^{d \times m}$ such that

$$\|B^T A z - z\|_\infty < \epsilon$$

for all k -sparse $z \in [-1,1]^m$. (Here, any bounded interval would suffice.)

Lean-expanded paper semantic target

type:

$\mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{R} \rightarrow \mathbb{N} \rightarrow \text{Prop}$

value:

```
fun (m k : ℕ) (epsilon : ℝ) (d : ℕ) =>
  GKP26LinearRepresentationHypothesis.linearCompressedSensingFeasible m k d epsilon ∧
  ∀ (d' : ℕ), GKP26LinearRepresentationHypothesis.linearCompressedSensingFeasible m k d' epsilon → d ≤ d'
```

Recorded source-to-declaration screening Verdict: matches

Reason: The source defines d as the smallest natural dimension for which some $A,B \in \mathbb{R}^{d \times m}$ recover every k -sparse $z \in [-1,1]^m$ with strict sup error below ϵ . The displayed Lean target is feasibility at d conjoined with $d \leq$ every feasible d' ; its displayed prerequisites expand feasibility to the two $\text{Fin-}m$ -indexed families of $\text{Fin-}d$ real columns, k -sparsity to support cardinality $\leq k$, the box to coordinatewise $|z_i| \leq 1$, and recovery to coordinatewise $|\sum_j z_j \langle B_i, A_j \rangle - z_i| < \epsilon$. Thus both the feasible set and its minimum are the source's ones.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

GKP26LinearRepresentationHypothesis.linearRecoveryCondition

Paper-local declaration:

GKP26LinearRepresentationHypothesis.linearRecoveryCondition

Source locator: cited publication, lines 205-209

Verbatim paper-source connection

Let $d(m,k,\epsilon)$ be the smallest choice of d such that there exists $A,B \in \mathbb{R}^{d \times m}$ such that

$\begin{equation}$

$\|B^T A z - z\|_\infty < \epsilon$

$\end{equation}$

for all k -sparse $z \in [-1,1]^m$. (Here, any bounded interval would suffice.)

[Next verbatim source excerpt]

$\subsubsection{Intuition and Proof Techniques}$

We begin by developing some geometric intuition for the problem. Let $a_1, \dots, a_m \in \mathbb{R}^d$ be the columns of $A \in \mathbb{R}^{d \times m}$ and

$\hookrightarrow$ $b_1, \dots, b_m \in \mathbb{R}^d$ be the columns of $B \in \mathbb{R}^{d \times m}$. For each $i \in [m]$, we recall that a_i is feature i 's $\textbf{representation}$

$\hookrightarrow$ $\textbf{vector}$ and b_i is feature i 's $\textbf{probe vector}$. Now let $\hat{z} = B^T A z$. We are interested in when

$\begin{equation} \label{eq:salmon}$

$\|\hat{z} - z\|_\infty < \epsilon \iff |\hat{z}_i - z_i| < \epsilon, \quad \forall i \in [m].$

$\end{equation}$

Note that

$\begin{equation}$

Lean-expanded paper semantic target

type:

$\{m\ d : \mathbb{N}\} \rightarrow$

$\text{GKP26LinearRepresentationHypothesis.FeatureMatrix } m\ d \rightarrow$

$\text{GKP26LinearRepresentationHypothesis.FeatureMatrix } m\ d \rightarrow \mathbb{N} \rightarrow \mathbb{R} \rightarrow \text{Prop}$

value:

$\text{fun } \{m\ d : \mathbb{N}\} (A\ B : \text{GKP26LinearRepresentationHypothesis.FeatureMatrix } m\ d) (k : \mathbb{N}) (\epsilon : \mathbb{R}) =>$

$\forall (z : \text{GKP26LinearRepresentationHypothesis.FeatureVector } m),$

$\text{GKP26LinearRepresentationHypothesis.kSparse } k\ z \rightarrow$

$\text{GKP26LinearRepresentationHypothesis.inUnitBox } z \rightarrow$

$\text{AppliedModelingLib.Math.LinearCompressedSensing.SuperErrorLt } A\ B\ z\ \epsilon$

Recorded source-to-declaration screening Verdict: matches

Reason: For fixed A and B , the source requires the error bound for every k -sparse $z \in [-1,1]^m$, and states that the infinity-norm inequality is equivalent to the strict coordinatewise inequalities for all feature indices. The Lean target has exactly that universal z quantifier and those two antecedents; `SuperErrorLt` and `linearProbe` expand the conclusion to $|\sum_j z_j < B_{i,A_j} - z_i| < \epsilon$ for every i . In the displayed feasibility use, the source's existential quantifiers over A and B are supplied outside this fixed-matrix condition, so no source quantifier is lost.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

GKP26LinearRepresentationHypothesis.paper_strict_mu_incoherent

Paper-local declaration:

GKP26LinearRepresentationHypothesis.paper_strict_mu_incoherent

Source locator: cited publication, lines 399-409

Verbatim paper-source connection

```
\begin{definition}[\mu$-incoherence]
  Say that a matrix $C$ with columns $c_1,\cdots,c_m$ is $\mu$-incoherent if
  \begin{equation}
    \langle c_i, c_i \rangle = 1
  \end{equation}
  for all $i \in [m]$, and
  \begin{equation}
    |\langle c_i, c_j \rangle| < \mu
  \end{equation}
  for all $i \neq j$.
\end{definition}
```

Lean-expanded paper semantic target

```
type:
{m d : ℕ} → GKP26LinearRepresentationHypothesis.paper_feature_matrix m d → ℝ → Prop

value:
fun {m d : ℕ} (A : GKP26LinearRepresentationHypothesis.paper_feature_matrix m d) (mu : ℝ) =>
  (∀ (i : Fin m), AppliedModelingLib.Math.LinearCompressedSensing.inner (A i) (A i) = 1) ∧
  ∀ {{i j : Fin m}}, i ≠ j → |AppliedModelingLib.Math.LinearCompressedSensing.inner (A i) (A j)| < mu
```

Recorded source-to-declaration screening Verdict: matches

Reason: The source requires every column c_i to have self-inner-product 1 and every distinct pair to satisfy $|\langle c_i, c_j \rangle| < \mu$. The Lean target quantifies the $\text{Fin } m$ -indexed columns of the displayed d -by- m paper matrix, uses the displayed finite real inner product over $\text{Fin } d$, and conjoins exactly the same normalization and strict off-diagonal bound under $i \neq j$.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Source claims

Review row 1

Source locator: cited publication, lines 184-199

Verbatim source input

Let us recall a classical result in compressed sensing \citep{candes2006near, candes2006robust, donoho2006compressed}, which exactly deals with the setting in which features are represented linearly but accessed non-linearly, answering Q1.

```
\begin{theorem}[Compressed Sensing]\label{thm:compressed-sensing}
  There exists a matrix  $A \in \mathbb{R}^{d \times m}$  with  $d = O\left(k \log \frac{m}{k}\right)$ 
  such that for all  $k$ -sparse  $z \in \mathbb{R}^m$ ,
  \begin{equation}
    g(x) := \operatorname{argmin}_{z' \in \mathbb{R}^m: x = Az'} \|z'\|_1
  \end{equation}
  satisfies
  \begin{equation}
    g(Az) = z.
  \end{equation}
\end{theorem}
```

This shows that when

```
\begin{equation}
  d = O\left(k \log \frac{m}{k}\right),
\end{equation}
```

Settled source reading Compressed sensing: the checked construction restricts the unstated sparse range to $1 \leq k$ and $2k \leq m$.

Expanded PaperInterface specification

```
∃ (C : ℝ),
  0 < C ∧
  ∀ (m k : ℕ),
    1 ≤ k →
      2 * k ≤ m →
        ∃ (d : ℕ) (A : GKP26LinearRepresentationHypothesis.paper_feature_matrix m d),
          ↑d ≤ C * ↑k * Real.log (↑m / ↑k) ∧
            GKP26LinearRepresentationHypothesis.paper_basis_pursuit_exact_recovery A k
```

Lean-elaborated claim roles

```
1. conclusion: ∃ (C : ℝ),
  0 < C ∧
  ∀ (m k : ℕ),
    1 ≤ k →
      2 * k ≤ m →
        ∃ (d : ℕ) (A : GKP26LinearRepresentationHypothesis.paper_feature_matrix m d),
          ↑d ≤ C * ↑k * Real.log (↑m / ↑k) ∧
            GKP26LinearRepresentationHypothesis.paper_basis_pursuit_exact_recovery A k
```

Lean proof endpoint (built separately)

GKP26LinearRepresentationHypothesis.paper_compressed_sensing_external_basis_pursuit

Recorded source-to-Spec screening Verdict: matches

Reason: Compared cited publication:184-199 with the expanded existential/universal target and all ten supplied prerequisites, using LRH-COMPRESSED-SENSING-SPARSE-DOMAIN-01 only for this row. The target chooses one positive constant C independent of m and k , then a natural dimension d and real d -by- m matrix for every $1 \leq k$ and $2k \leq m$, with $d \leq C * k * \log(m/k)$, precisely the approved nontrivial sparse domain of the displayed O rate. The matrix/vector aliases give real feature-indexed columns and real feature vectors; the measurement sum is Az and l_1 is the sum of coordinate absolute values. The sparse predicate imposes support cardinality at most k . Expanding both recovery wrappers quantifies over every such sparse z and every real competitor z' with $Az' = Az$: $l_1(z') \leq l_1(z)$ forces $z' = z$. Since z itself is feasible, this says z is the unique l_1 minimizer at Az , matching $g(Az) = z$ for the displayed argmin. No extra restriction on coefficients or competing vectors appears, and the approval is a source-regime clarification, not a materially corrected target.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Source locator: cited publication, lines 547-610

Verbatim source input

```
\begin{proposition}\label{prop:geo}
Consider constants  $\epsilon, \delta$ 
\begin{equation}
```

$$\begin{aligned}
& \text{end{equation}} \\
& \text{for all } \$k\$-sparse \$z\$ in $[-1,1]^m$, and where \\
& \begin{aligned}
& \text{begin{enumerate}} \\
& \quad \text{item{(i)}} \$\angle \frac{\{a_i\}}{\{a_i\}}, \frac{\{b_i\}}{\{b_i\}} \angle \delta + o(1)$ for all \$i\$ in $[m]$, \\
& \quad \text{item{(ii)}} \$\angle \frac{\{a_i\}}{\{a_i\}}, \frac{\{a_i\}}{\{a_i\}} \angle \\
& \quad > 1 - \delta - o(1)$ for all \$i \neq j$, \\
& \quad \text{item{(iii)}} \$\angle \frac{\{b_i\}}{\{b_i\}}, \frac{\{b_i\}}{\{b_i\}} \angle > 1 - \delta - o(1)$ for all \$i \neq j$. \\
& \text{end{enumerate}} \\
& \text{end{proposition}} \\
& \\
& \text{begin{proof}} \\
& \text{We give a construction. Let } \$c_1, c_2, \dots, c_m, a^*, b^*$ be the columns of a μ-incoherent matrix $\mathcal{R}^{\mathcal{R}} \in \mathbb{R}^{d \times (m+2)}$. Then set \\
& \begin{aligned}
& \text{begin{align}} \\
& \quad a_i = c_i + \lambda a^* \\
& \quad b_i = c_i + \lambda b^* \\
& \text{end{align}} \\
& \text{for } \lambda = \sqrt{\frac{1}{\delta} - 1}. \\
& \text{We begin by showing that for an appropriately chosen } \μ, $\|B - z\|_\infty < \epsilon$ for all \$k\$-sparse \$z\$ in $[-1,1]^m$. Now let $\hat{z}$ \\
& \hookrightarrow = B - z$ for any z in $[-1,1]^m$ with support T. Then \\
& \begin{aligned}
& \text{begin{equation}} \\
& \quad \hat{z}_i = \left| \angle b_i, \sum_{j=1}^m z_j a_j \right| \\
& \quad = z_i \angle b_i, a_i + \sum_{j \in T \setminus \{i\}} z_j \angle b_i, a_j \\
& \text{end{equation}} \\
& \text{Therefore,} \\
& \begin{aligned}
& \text{begin{align}} \\
& \quad |\hat{z}_i - z_i| \leq |z_i (\angle b_i, a_i - 1)| + \sum_{j \in T \setminus \{i\}} |z_j \angle b_i, a_j| \\
& \quad \leq |z_i (\angle b_i, a_i - 1)| + \sum_{j \in T \setminus \{i\}} |\angle b_i, a_j| \\
& \text{end{align}} \\
& \text{where the last inequality follows from observing that } $z_j \in [-1,1]$ for all $j \in [m]$. Now observe that \\
& \begin{aligned}
& \text{begin{align}} \\
& \quad |z_i (\angle b_i, a_i - 1)| \leq |z_i| |\angle c_i + \lambda b^*, c_i + \lambda a^* - 1| \\
& \quad \leq |z_i| (|\angle c_i, \lambda b^*| + |\angle \lambda b^*, c_i| + |\angle \lambda b^*, \lambda a^*|) \\
& \quad \hookrightarrow \text{label{eq:apple-1}} \\
& \quad \leq |z_i| (2\lambda + \lambda^2) \mu \text{label{eq:apple-2}} \\
& \quad \leq |z_i| (1 + \lambda)^2 \mu, \\
& \text{end{align}} \\
& \text{where \text{label{eq:apple-1}}} follows because $|\angle c_i, c_i| = 1$ and \text{label{eq:apple-2}} follows from μ-incoherence. Similarly, \\
& \begin{aligned}
& \text{begin{align}} \\
& \quad |\angle b_i, a_j| \leq |\angle c_i + \lambda b^*, c_i + \lambda a^*| \\
& \quad \leq |\angle c_i, c_j| + |\angle c_i, \lambda a^*| + |\angle \lambda b^*, c_i| + |\angle \lambda b^*, \lambda a^*| \\
& \quad \leq (1 + 2\lambda + \lambda^2) \mu = (1 + \lambda)^2 \mu. \\
& \text{end{align}} \\
& \text{This implies that} \\
& \begin{aligned}
& \text{begin{equation}} \\
& \quad |z_i (\angle b_i, a_i - 1)| + \sum_{j \in T \setminus \{i\}} |\angle b_i, a_j| \leq k (1 + \lambda)^2 \mu, \\
& \text{end{equation}} \\
& \text{where we use the fact that if } $|z_j| > 0$, then $j \in T$. Therefore, it suffices to take $\mu = \frac{\epsilon}{k(1+\lambda)^2}$ to \\
& \hookrightarrow ensure that $\|B - z\|_\infty < \epsilon$. From \text{prop:incoherent}, there exists a μ-incoherent matrix in $\mathcal{R}^{\mathcal{R}} \in \mathbb{R}^{d \times (m+2)}$ \\
& \text{for} \\
& \begin{aligned}
& \text{begin{equation}} \\
& \quad d = O(k^2 (1 + \lambda)^4 \epsilon^2 \log(m+2)) = O(k^2 (1 + \lambda)^4 \epsilon^2 \log m) \\
& \quad \hookrightarrow O(\epsilon, \delta) (k^2 \log m). \\
& \text{end{equation}} \\
& \\
& \text{paragraph{Establishing (i).}} We now show that the construction satisfies (i). Observe that for all $i \in [m]$, \\
& \begin{aligned}
& \text{begin{align}} \\
& \quad \left| \angle \frac{\{a_i\}}{\{a_i\}}, \frac{\{b_i\}}{\{b_i\}} \right| \\
& \quad \leq \frac{|\angle c_i + \lambda b^*, c_i + \lambda a^*|}{|\angle c_i, c_i| + |\angle c_i, \lambda a^*| + |\angle \lambda b^*, c_i| + |\angle \lambda b^*, \lambda a^*|} \\
& \quad \hookrightarrow \frac{|\angle c_i, c_i| + |\angle c_i, \lambda a^*| + |\angle \lambda b^*, c_i| + |\angle \lambda b^*, \lambda a^*|}{|\angle c_i, c_i| + |\angle c_i, \lambda a^*| + |\angle \lambda b^*, c_i| + |\angle \lambda b^*, \lambda a^*|} \\
& \quad \leq \frac{1 + (2\lambda + \lambda^2) \mu}{1 + \lambda} = \frac{1 + \frac{2\lambda + \lambda^2}{k(1+\lambda)^2} \epsilon}{1 + \lambda} = \frac{1 + \frac{2\lambda + \lambda^2}{k(1+\lambda)^2} \epsilon}{1 + \lambda} \\
& \quad \hookrightarrow \frac{1 + \frac{2\lambda + \lambda^2}{k(1+\lambda)^2} \epsilon}{1 + \lambda} = \frac{1 + \frac{2\lambda + \lambda^2}{k(1+\lambda)^2} \epsilon}{1 + \lambda} \\
& \text{end{align}} \\
& \text{Note that} \\
& \begin{aligned}
& \text{begin{align}} \\
& \quad \angle a_i, a_i \leq \angle c_i + \lambda a^*, c_i + \lambda a^* \leq \angle c_i, c_i + 2\angle c_i, \lambda a^* + \angle \lambda a^*, \lambda a^* \\
& \quad \hookrightarrow \angle \lambda a^*, \lambda a^* \\
& \quad \leq 1 + \lambda^2 - \frac{2\lambda \epsilon}{k(1+\lambda)^2} = 1 + \lambda^2 - o(1). \\
& \text{end{align}} \\
& \end{aligned}
\end{aligned}$$


```

(AppliedModelingLib.Math.TendsToZero fun (k : ℕ) =>
  1 - delta -
  GKP26LinearRepresentationHypothesis.geometry1OffdiagCorrelationEnvelope
    (GKP26LinearRepresentationHypothesis.geometry1SourceLambda delta)
    (GKP26LinearRepresentationHypothesis.geometry1SourceMu epsilon delta k)) ∧
  ∀ (m k : ℕ),
  ∃ (D : ℕ),
  ↑D ≤
    GKP26LinearRepresentationHypothesis.geometry1SourceDimensionCoefficient epsilon delta *
    ↑(k + 1) ^ 2 *
    Real.log (2 * ↑(m + 2) ^ 2) +
    2 ^
  ∃ (A : GKP26LinearRepresentationHypothesis.paper_feature_matrix m D) (B :
    GKP26LinearRepresentationHypothesis.paper_feature_matrix m D),
    GKP26LinearRepresentationHypothesis.linearRecoveryCondition A B k epsilon ∧
    (∀ (i : Fin m),
      |GKP26LinearRepresentationHypothesis.columnCorrelation (A i) (B i)| ≤
        GKP26LinearRepresentationHypothesis.geometry1SelfCorrelationEnvelope
          (GKP26LinearRepresentationHypothesis.geometry1SourceLambda delta)
          (GKP26LinearRepresentationHypothesis.geometry1SourceMu epsilon delta k)) ∧
    (∀ {i j : Fin m},
      i ≠ j →
        GKP26LinearRepresentationHypothesis.geometry1OffdiagCorrelationEnvelope
          (GKP26LinearRepresentationHypothesis.geometry1SourceLambda delta)
          (GKP26LinearRepresentationHypothesis.geometry1SourceMu epsilon delta k) ≤
            GKP26LinearRepresentationHypothesis.columnCorrelation (A i) (A j)) ∧
    (∀ {i j : Fin m},
      i ≠ j →
        GKP26LinearRepresentationHypothesis.geometry1OffdiagCorrelationEnvelope
          (GKP26LinearRepresentationHypothesis.geometry1SourceLambda delta)
          (GKP26LinearRepresentationHypothesis.geometry1SourceMu epsilon delta k) ≤
            GKP26LinearRepresentationHypothesis.columnCorrelation (B i) (B j))

```

Lean-elaborated claim roles

1. parameter: $\mathbb{R}$
2. parameter: $\mathbb{R}$
3. assumption: $0 < \epsilon$
4. assumption: $\epsilon < 1$
5. assumption: $0 < \delta$
6. assumption: $\delta < 1$
7. conclusion: (AppliedModelingLib.Math.TendsToZero fun (k : ℕ) =>
 GKP26LinearRepresentationHypothesis.geometry1SelfCorrelationEnvelope
 (GKP26LinearRepresentationHypothesis.geometry1SourceLambda delta)
 (GKP26LinearRepresentationHypothesis.geometry1SourceMu epsilon delta k) -
 delta) ∧
 (AppliedModelingLib.Math.TendsToZero fun (k : ℕ) =>
 1 - delta -
 GKP26LinearRepresentationHypothesis.geometry1OffdiagCorrelationEnvelope
 (GKP26LinearRepresentationHypothesis.geometry1SourceLambda delta)
 (GKP26LinearRepresentationHypothesis.geometry1SourceMu epsilon delta k)) ∧
 ∀ (m k : ℕ),
 ∃ (D : ℕ),
 ↑D ≤
 GKP26LinearRepresentationHypothesis.geometry1SourceDimensionCoefficient epsilon delta * ↑(k + 1) ^ 2 *
 Real.log (2 * ↑(m + 2) ^ 2) +
 2 ^
 ∃ (A : GKP26LinearRepresentationHypothesis.paper_feature_matrix m D) (B :
 GKP26LinearRepresentationHypothesis.paper_feature_matrix m D),
 GKP26LinearRepresentationHypothesis.linearRecoveryCondition A B k epsilon ∧
 (∀ (i : Fin m),
 |GKP26LinearRepresentationHypothesis.columnCorrelation (A i) (B i)| ≤
 GKP26LinearRepresentationHypothesis.geometry1SelfCorrelationEnvelope
 (GKP26LinearRepresentationHypothesis.geometry1SourceLambda delta)
 (GKP26LinearRepresentationHypothesis.geometry1SourceMu epsilon delta k)) ∧
 (∀ {i j : Fin m},
 i ≠ j →
 GKP26LinearRepresentationHypothesis.geometry1OffdiagCorrelationEnvelope
 (GKP26LinearRepresentationHypothesis.geometry1SourceLambda delta)
 (GKP26LinearRepresentationHypothesis.geometry1SourceMu epsilon delta k) ≤
 GKP26LinearRepresentationHypothesis.columnCorrelation (A i) (A j)) ∧
 (∀ {i j : Fin m},
 i ≠ j →
 GKP26LinearRepresentationHypothesis.geometry1OffdiagCorrelationEnvelope
 (GKP26LinearRepresentationHypothesis.geometry1SourceLambda delta)
 (GKP26LinearRepresentationHypothesis.geometry1SourceMu epsilon delta k) ≤
 GKP26LinearRepresentationHypothesis.columnCorrelation (B i) (B j))

Lean proof endpoint (built separately)

GKP26LinearRepresentationHypothesis.paper_geometry1_source_constructed_rate

Recorded source-to-Spec screening Verdict: matches

Reason: The source proposition at cited publication:547-610 assumes $0 < \epsilon, \delta < 1$ and constructs $d = O(\epsilon, \delta)(k^2 \log m)$ matrices with uniform k -sparse box recovery, same-column correlation $\delta + o(1)$, and off-diagonal correlations $1 - \delta - o(1)$. The expanded target gives an explicit finite O -envelope with such matrices and recovery, while the two TendsToZero conjuncts and the displayed correlation envelopes encode exactly those three asymptotic correlation conclusions.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 3

Source locator: cited publication, lines 638-680

Verbatim source input

```
\begin{proposition}\label{prop:geometry-2}
Consider any constants  $\epsilon > 0$ ,  $\gamma \geq 1$ . Then if  $\|a_i\|, \|b_i\| \leq \gamma$  for all  $i \in [m]$  and
\begin{equation}
\|B^T A z - z\|_\infty < \epsilon
\end{equation}
for all  $k$ -sparse  $z \in [-1, 1]^m$ , it holds that
\begin{enumerate}
\item(i)  $\angle \frac{a_i}{\|a_i\|}, \frac{b_i}{\|b_i\|} \geq \frac{1-\epsilon}{\gamma^2}$  for all  $i \in [m]$ ,
\item(ii)  $\angle \frac{a_i}{\|a_i\|}, \frac{a_j}{\|a_j\|} \geq \frac{\epsilon}{\gamma^2} \frac{1-\epsilon}{1-\epsilon} + \sqrt{1-\frac{\epsilon}{\gamma^2}}$ 
 $\hookrightarrow \frac{\epsilon}{\gamma^2} \frac{1-\epsilon}{\gamma^4}$  for all  $i \neq j$ ,
\item(iii)  $\angle \frac{b_i}{\|b_i\|}, \frac{b_j}{\|b_j\|} \geq \frac{\epsilon}{\gamma^2} \frac{1-\epsilon}{1-\epsilon} + \sqrt{1-\frac{\epsilon}{\gamma^2}}$ 
 $\hookrightarrow \frac{\epsilon}{\gamma^2} \frac{1-\epsilon}{\gamma^4}$  for all  $i \neq j$ .
\end{enumerate}
\end{proposition}

\begin{proof}
We must have that  $\angle a_i, b_i \geq 1 - \epsilon$  since  $\|B^T A e_i - e_i\|_\infty \leq \epsilon$ . It follows directly that
\begin{equation}
\angle a_i, b_i \geq 1 - \epsilon \implies \left| \angle \frac{a_i}{\|a_i\|}, \frac{b_i}{\|b_i\|} \right| \geq \frac{1-\epsilon}{\gamma^2},
\end{equation}
establishing (i).

We now show (ii). By Cauchy-Schwarz,  $\|a_i\| \|b_i\| \geq \angle a_i, b_i \geq 1 - \epsilon$  for all  $i \in [m]$ . So since
 $\hookrightarrow \|a_i\| \|b_i\| \leq \gamma^2$  for all  $i \in [m]$ , we have that  $\|a_i\|, \|b_i\| \geq \frac{1-\epsilon}{\gamma^2}$  for all  $i \in [m]$ .
 $\hookrightarrow$  Therefore, we have that
\begin{equation}
\left| \angle \frac{b_i}{\|b_i\|}, \frac{a_j}{\|a_j\|} \right| \leq \angle b_i, a_j \leq \frac{\epsilon}{\gamma^2} \frac{1-\epsilon}{1-\epsilon} <
\hookrightarrow \frac{\epsilon}{\gamma^2} \frac{1-\epsilon}{\gamma^2}.
\end{equation}
Let  $\tilde{a}_i$  denote  $\frac{a_i}{\|a_i\|}$  and  $\tilde{b}_i$  denote  $\frac{b_i}{\|b_i\|}$  for all  $i \in [m]$ . Then write
\begin{align}
\tilde{a}_i &= \alpha_i \tilde{b}_i + \tilde{a}_i^\perp \\
\tilde{a}_j &= \alpha_j \tilde{b}_i + \tilde{a}_j^\perp,
\end{align}
where  $\alpha_i = \angle \tilde{a}_i, \tilde{b}_i$ ,  $\alpha_j = \angle \tilde{a}_j, \tilde{b}_i$ , and  $\tilde{a}_i^\perp, \tilde{a}_j^\perp \perp \tilde{b}_i$ . Then
\begin{equation}
\angle \tilde{a}_i, \tilde{a}_j = \angle \alpha_i \tilde{b}_i + \alpha_j \tilde{a}_i^\perp, \tilde{a}_j^\perp \geq \angle \tilde{a}_i^\perp, \tilde{a}_j^\perp.
\end{equation}
By Cauchy-Schwarz,
\begin{equation}
|\angle \tilde{a}_i^\perp, \tilde{a}_j^\perp| \leq \|\tilde{a}_i^\perp\| \|\tilde{a}_j^\perp\| = \sqrt{(1-\alpha_i^2)(1-\alpha_j^2)}.
\end{equation}
Therefore,
\begin{equation}
|\angle \tilde{a}_i, \tilde{a}_j| \leq |\alpha_i| |\alpha_j| + \sqrt{(1-\alpha_i^2)(1-\alpha_j^2)}.
\end{equation}
We have that  $\alpha_i \geq \frac{1-\epsilon}{\gamma^2}$  and  $0 < \alpha_j < \frac{\epsilon}{\gamma^2} \frac{1-\epsilon}{1-\epsilon}$ . Therefore,
\begin{equation}
```

Settled source reading Geometry Proposition 2: $\epsilon > 0$ becomes $0 < \epsilon < 1$; the first denominator in parts (ii)/(iii) changes from $1 - \epsilon$ to $(1 - \epsilon)^2$, giving a weaker finite bound.

Formalized review target The archival source above is not asserted equivalent to this formalized target.

Under the stated recovery and column-norm hypotheses, Geometry Proposition 2(iii) holds with first term $\epsilon * \gamma^2 / (1-\epsilon)^2$ in the $\hookrightarrow$ normalized probe-vector correlation bound.

Archival source anchor: cited publication, lines 638-680

Expanded PaperInterface specification

```
∀ {m d k : ℕ} {epsilon gamma : ℝ} {A B : GKP26LinearRepresentationHypothesis.paper_feature_matrix m d},
1 ≤ k →
0 < epsilon →
epsilon < 1 →
1 ≤ gamma →
GKP26LinearRepresentationHypothesis.linearRecoveryCondition A B k epsilon →
(∀ (i : Fin m), GKP26LinearRepresentationHypothesis.paper_column_norm (A i) ≤ gamma) →
(∀ (i : Fin m), GKP26LinearRepresentationHypothesis.paper_column_norm (B i) ≤ gamma) →
∀ {i j : Fin m},
i ≠ j →
GKP26LinearRepresentationHypothesis.paper_column_correlation (B i) (B j) ≤
epsilon * gamma ^ 2 / (1 - epsilon) ^ 2 + √(1 - ((1 - epsilon) / gamma ^ 2) ^ 2)
```

Lean-elaborated claim roles

1. parameter: $\mathbb{N}$
2. parameter: $\mathbb{N}$
3. parameter: $\mathbb{N}$
4. parameter: $\mathbb{R}$
5. parameter: $\mathbb{R}$
6. parameter: GKP26LinearRepresentationHypothesis.paper_feature_matrix m d
7. parameter: GKP26LinearRepresentationHypothesis.paper_feature_matrix m d
8. assumption: $1 \leq k$
9. assumption: $0 < \epsilon$
10. assumption: $\epsilon < 1$
11. assumption: $1 \leq \gamma$
12. assumption: GKP26LinearRepresentationHypothesis.linearRecoveryCondition A B k epsilon
13. assumption: $\forall (i : \text{Fin } m), \text{GKP26LinearRepresentationHypothesis.paper_column_norm } (A \ i) \leq \gamma$
14. assumption: $\forall (i : \text{Fin } m), \text{GKP26LinearRepresentationHypothesis.paper_column_norm } (B \ i) \leq \gamma$
15. parameter: $\text{Fin } m$
16. parameter: $\text{Fin } m$
17. assumption: $i \neq j$
18. conclusion: $\text{Real.le} \uparrow (\text{GKP26LinearRepresentationHypothesis.paper_column_correlation } (B \ i) \ (B \ j))$
 $(\epsilon * \gamma^2 / (1 - \epsilon))^2 + \sqrt{1 - ((1 - \epsilon) / \gamma^2)^2}$

Lean proof endpoint (built separately)

GKP26LinearRepresentationHypothesis.paper_geometry2_probe_column_correlation_upper_corrected

Recorded source-to-Spec screening Verdict: matches approved corrected target

Reason: For cited publication:638-680, approved context LRH-GEOMETRY2-NONDEGENERATE-EPSILON-01 requires $0 < \epsilon < 1$ and $\gamma \geq 1$, and the expanded target has exactly those premises together with recovery, both column-norm bounds, and $i \neq j$. Its $B \ i, B \ j$ conclusion uses $\epsilon * \gamma^2 / (1 - \epsilon)^2$ plus $\sqrt{1 - ((1 - \epsilon) / \gamma^2)^2}$, exactly the disclosed LRH-GEOMETRY2-DENOMINATOR-01 corrected target for Proposition 2(iii), whose complete approval record is displayed.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 4

Source locator: cited publication, lines 638-680

Verbatim source input

```
\begin{proposition}\label{prop:geometry-2}
Consider any constants  $\epsilon > 0$ ,  $\gamma \geq 1$ . Then if  $\|a_i\|, \|b_i\| \leq \gamma$  for all  $i \in [m]$  and
\begin{equation}
\|B^{\top}Az - z\|_{\infty} < \epsilon
\end{equation}
for all  $k$ -sparse  $z \in [-1, 1]^m$ , it holds that
\begin{enumerate}
\item(i)  $\angle \frac{a_i}{\|a_i\|}, \frac{b_i}{\|b_i\|} \geq \frac{1-\epsilon}{\gamma^2}$  for all  $i \in [m]$ ,
\item(ii)  $\angle \frac{a_i}{\|a_i\|}, \frac{a_j}{\|a_j\|}, \frac{b_i}{\|b_i\|} \geq \frac{1-\epsilon}{\gamma^2} + \sqrt{1 - \frac{1-\epsilon}{\gamma^2}}$  for all  $i \neq j$ ,
\item(iii)  $\angle \frac{b_i}{\|b_i\|}, \frac{b_j}{\|b_j\|} \geq \frac{1-\epsilon}{\gamma^2} + \sqrt{1 - \frac{1-\epsilon}{\gamma^2}}$  for all  $i \neq j$ .
\end{enumerate}
\end{proposition}

\begin{proof}
We must have that  $\angle a_i, b_i \geq 1 - \epsilon$  since  $\|B^{\top}Ae_i - e_i\|_{\infty} \leq \epsilon$ . It follows directly that
\begin{equation}
\angle a_i, b_i > 1 - \epsilon \implies \left| \angle \frac{a_i}{\|a_i\|}, \frac{b_i}{\|b_i\|} \right| \geq \frac{1-\epsilon}{\gamma^2},
\end{equation}
establishing (i).

We now show (ii). By Cauchy-Schwarz,  $\|a_i\| \|b_i\| \geq \angle a_i, b_i > 1 - \epsilon$  for all  $i \in [m]$ . So since
 $\|a_i\| \|b_i\| \leq \lambda$  for all  $i \in [m]$ , we have that  $\|a_i\|, \|b_i\| \geq \frac{1-\epsilon}{\gamma^2}$  for all  $i \in [m]$ .
Therefore, we have that
\begin{equation}
\left| \angle \frac{b_i}{\|b_i\|}, \frac{a_j}{\|a_j\|} \right| \leq \angle b_i, a_j \leq \frac{1-\epsilon}{\gamma^2} + \sqrt{1 - \frac{1-\epsilon}{\gamma^2}}.
\end{equation}
Let  $\tilde{a}_i$  denote  $\frac{a_i}{\|a_i\|}$  and  $\tilde{b}_i$  denote  $\frac{b_i}{\|b_i\|}$  for all  $i \in [m]$ . Then write
\begin{align}
\tilde{a}_i &= \alpha_i \tilde{b}_i + \tilde{a}_i_{\perp} \\
\tilde{a}_j &= \alpha_j \tilde{b}_i + \tilde{a}_j_{\perp},
\end{align}
where  $\alpha_i = \angle \tilde{a}_i, \tilde{b}_i$ ,  $\alpha_j = \angle \tilde{a}_j, \tilde{b}_i$ , and  $\tilde{a}_i_{\perp}, \tilde{a}_j_{\perp} \perp \tilde{b}_i$ . Then
\begin{equation}
\angle \tilde{a}_i, \tilde{a}_j = \angle \alpha_i \tilde{b}_i + \alpha_j \tilde{b}_i, \tilde{a}_i_{\perp}, \tilde{a}_j_{\perp} \leq \angle \tilde{a}_i_{\perp}, \tilde{a}_j_{\perp}.
\end{equation}
By Cauchy-Schwarz,
\begin{equation}
|\angle \tilde{a}_i_{\perp}, \tilde{a}_j_{\perp}| \leq \|\tilde{a}_i_{\perp}\| \|\tilde{a}_j_{\perp}\| \leq \sqrt{(1 - \alpha_i^2)(1 - \alpha_j^2)}.
\end{equation}
Therefore,
\begin{equation}
|\angle \tilde{a}_i, \tilde{a}_j| \leq |\alpha_i| |\alpha_j| + \sqrt{(1 - \alpha_i^2)(1 - \alpha_j^2)}.
\end{equation}
We have that  $\alpha_i \geq \frac{1-\epsilon}{\gamma^2}$  and  $0 < \alpha_j < \frac{1-\epsilon}{\gamma^2} + \sqrt{1 - \frac{1-\epsilon}{\gamma^2}}$ . Therefore,
\begin{equation}
```

Settled source reading Geometry Proposition 2: $\epsilon > 0$ becomes $0 < \epsilon < 1$; the first denominator in parts (ii)/(iii) changes from $1 - \epsilon$ to $(1 - \epsilon)^2$, giving a weaker finite bound.

Formalized review target The archival source above is not asserted equivalent to this formalized target.

Under the stated recovery and column-norm hypotheses, Geometry Proposition 2(ii) holds with first term $\epsilon * \gamma^2 / (1 - \epsilon)^2$ in the normalized representation-vector correlation bound.

Archival source anchor: cited publication, lines 638-680

Expanded PaperInterface specification

```
∀ {m d k : ℕ} {epsilon gamma : ℝ} {A B : GKP26LinearRepresentationHypothesis.paper_feature_matrix m d},
1 ≤ k →
0 < epsilon →
epsilon < 1 →
1 ≤ gamma →
GKP26LinearRepresentationHypothesis.linearRecoveryCondition A B k epsilon →
(∀ (i : Fin m), GKP26LinearRepresentationHypothesis.paper_column_norm (A i) ≤ gamma) →
(∀ (i : Fin m), GKP26LinearRepresentationHypothesis.paper_column_norm (B i) ≤ gamma) →
∀ {i j : Fin m},
i ≠ j →
GKP26LinearRepresentationHypothesis.paper_column_correlation (A i) (A j) ≤
epsilon * gamma ^ 2 / (1 - epsilon) ^ 2 + √(1 - ((1 - epsilon) / gamma ^ 2) ^ 2)
```

Lean-elaborated claim roles

1. parameter: $\mathbb{N}$
 2. parameter: $\mathbb{N}$
 3. parameter: $\mathbb{N}$
 4. parameter: $\mathbb{R}$
 5. parameter: $\mathbb{R}$
 6. parameter: GKP26LinearRepresentationHypothesis.paper_feature_matrix m d
 7. parameter: GKP26LinearRepresentationHypothesis.paper_feature_matrix m d
 8. assumption: $1 \leq k$
 9. assumption: $0 < \epsilon$
 10. assumption: $\epsilon < 1$
 11. assumption: $1 \leq \gamma$
 12. assumption: GKP26LinearRepresentationHypothesis.linearRecoveryCondition A B k epsilon
 13. assumption: $\forall (i : \text{Fin } m), \text{GKP26LinearRepresentationHypothesis.paper_column_norm } (A \ i) \leq \gamma$
 14. assumption: $\forall (i : \text{Fin } m), \text{GKP26LinearRepresentationHypothesis.paper_column_norm } (B \ i) \leq \gamma$
 15. parameter: $\text{Fin } m$
 16. parameter: $\text{Fin } m$
 17. assumption: $i \neq j$
 18. conclusion: $\text{Real.le} \uparrow (\text{GKP26LinearRepresentationHypothesis.paper_column_correlation } (A \ i) \ (A \ j))$
 $(\epsilon * \gamma^2 / (1 - \epsilon)^2 + \sqrt{1 - ((1 - \epsilon) / \gamma^2)^2})$

Lean proof endpoint (built separately)

GKP26LinearRepresentationHypothesis.paper_geometry2_representation_column_correlation_upper_corrected

Recorded source-to-Spec screening Verdict: matches approved corrected target

Reason: For cited publication:638-680, approved context LRH-GEOMETRY2-NONDEGENERATE-EPSILON-01 requires $0 < \epsilon < 1$ and $\gamma \geq 1$, and the expanded target has exactly those premises together with recovery, both column-norm bounds, and $i \neq j$. Its $A \ i, A \ j$ conclusion uses $\epsilon * \gamma^2 / (1 - \epsilon)^2$ plus $\sqrt{1 - ((1 - \epsilon) / \gamma^2)^2}$, exactly the disclosed LRH-GEOMETRY2-DENOMINATOR-01 corrected target for Proposition 2(ii), whose complete approval record is displayed.

Reviewer check: ☐ Matches formalized target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 5

Source locator: cited publication, lines 638-655

Verbatim source input

```
\begin{proposition}\label{prop:geometry-2}
Consider any constants  $\epsilon > 0$ ,  $\gamma \geq 1$ . Then if  $\|a_i\|, \|b_i\| \leq \gamma$  for all  $i \in [m]$  and
\begin{equation}
\|B^T A z - z\|_\infty < \epsilon
\end{equation}
for all  $k$ -sparse  $z \in [-1, 1]^m$ , it holds that
\begin{enumerate}
\item(i)  $\angle \frac{a_i}{\|a_i\|}, \frac{b_i}{\|b_i\|} \geq \frac{1-\epsilon}{\gamma^2}$  for all  $i \in [m]$ ,
\item(ii)  $\angle \frac{a_i}{\|a_i\|}, \frac{a_j}{\|a_j\|} \leq \frac{\epsilon}{\gamma^2} + \sqrt{1 - \frac{1-\epsilon}{\gamma^2}}$ 
 $\Rightarrow \epsilon^2 \gamma^4$  for all  $i \neq j$ ,
\item(iii)  $\angle \frac{b_i}{\|b_i\|}, \frac{b_j}{\|b_j\|} \leq \frac{\epsilon}{\gamma^2} + \sqrt{1 - \frac{1-\epsilon}{\gamma^2}}$ 
 $\Rightarrow \epsilon^2 \gamma^4$  for all  $i \neq j$ .
\end{enumerate}
\end{proposition}

\begin{proof}
We must have that  $\angle a_i, b_i \geq 1 - \epsilon$  since  $\|B^T A e_i - e_i\|_\infty \leq \epsilon$ . It follows directly that
\begin{equation}
\angle a_i, b_i > 1 - \epsilon \implies \left| \angle \frac{a_i}{\|a_i\|}, \frac{b_i}{\|b_i\|} \right| \leq \frac{1-\epsilon}{\gamma^2},
\end{equation}
```

Settled source reading Geometry Proposition 2: $\epsilon > 0$ becomes $0 < \epsilon < 1$; the first denominator in parts (ii)/(iii) changes from $1 - \epsilon$ to $(1 - \epsilon)^2$, giving a weaker finite bound.

Expanded PaperInterface specification

```
∀ {m d k : ℕ} {epsilon gamma : ℝ} {A B : GKP26LinearRepresentationHypothesis.paper_feature_matrix m d},
1 ≤ k →
0 < epsilon →
epsilon < 1 →
1 ≤ gamma →
GKP26LinearRepresentationHypothesis.linearRecoveryCondition A B k epsilon →
(∀ (i : Fin m), GKP26LinearRepresentationHypothesis.paper_column_norm (A i) ≤ gamma) →
(∀ (i : Fin m), GKP26LinearRepresentationHypothesis.paper_column_norm (B i) ≤ gamma) →
∀ (i : Fin m),
(1 - epsilon) / gamma ^ 2 ≤ GKP26LinearRepresentationHypothesis.paper_column_correlation (B i) (A i)
```

Lean-elaborated claim roles

1. parameter: $\mathbb{N}$
2. parameter: $\mathbb{N}$
3. parameter: $\mathbb{N}$
4. parameter: $\mathbb{R}$
5. parameter: $\mathbb{R}$
6. parameter: $\text{GKP26LinearRepresentationHypothesis.paper_feature_matrix } m \ d$
7. parameter: $\text{GKP26LinearRepresentationHypothesis.paper_feature_matrix } m \ d$
8. assumption: $1 \leq k$
9. assumption: $0 < \epsilon$
10. assumption: $\epsilon < 1$
11. assumption: $1 \leq \gamma$
12. assumption: $\text{GKP26LinearRepresentationHypothesis.linearRecoveryCondition } A \ B \ k \ \epsilon$
13. assumption: $\forall (i : \text{Fin } m), \text{GKP26LinearRepresentationHypothesis.paper_column_norm } (A \ i) \leq \gamma$
14. assumption: $\forall (i : \text{Fin } m), \text{GKP26LinearRepresentationHypothesis.paper_column_norm } (B \ i) \leq \gamma$
15. parameter: $\text{Fin } m$
16. conclusion: $\text{Real.le } ((1 - \epsilon) / \gamma ^ 2) (\text{GKP26LinearRepresentationHypothesis.paper_column_correlation } (B \ i) (A \ i))$

Lean proof endpoint (built separately)

$\text{GKP26LinearRepresentationHypothesis.paper_geometry2_self_column_correlation_lowerSpec_proof}$

Recorded source-to-Spec screening Verdict: matches

Reason: The archival Proposition 2(i) source at cited publication:638-655 and approved context LRH-GEOMETRY2-NONDEGENERATE-EPSILON-01 require $0 < \epsilon < 1$, $\gamma \geq 1$, uniform recovery, and both column-norm bounds. The expanded target has those premises and concludes for every i the same normalized self-correlation lower bound $(1-\epsilon)/\gamma^2$.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 6

Source locator: cited publication, lines 220-227

Verbatim source input

```
\begin{theorem}[Lower Bound]\label{thm:lower}
For  $\epsilon > \frac{k^{\frac{3}{2}}\sqrt{5}}{\sqrt{m}}$ ,
\begin{equation}
d(m,k,\epsilon) = \Omega_{\epsilon} \left( \frac{k^2}{\log k} \log \frac{m}{k} \right).
\end{equation}
\end{theorem}
```

(Here, the restriction $\epsilon > \frac{k^{\frac{3}{2}}\sqrt{5}}{\sqrt{m}}$ is not very consequential, since we are generally interested in the regime where $k \rightarrow \infty$ and $m \rightarrow \infty$.)

[Next verbatim source excerpt]

[Upper bound sketch.] We can give a fairly straightforward construction to establish the upper bound. In particular, we can give a construction where $A=B$. Then it suffices to find a matrix A such that $\langle a_i, a_j \rangle = 1$ and such that $\langle a_i, a_j \rangle < \frac{\epsilon}{k}$. Leveraging the k -sparsity of A , this would show that the total interference for any feature is bounded by $k \cdot \frac{\epsilon}{k} = \epsilon$. We can then use standard constructions of incoherent matrices (i.e., matrices with approximately-orthonormal columns), such as appropriately-scaled random Gaussian matrices, to obtain the desired bound.

[Initial attempt at lower bound.] To prove the lower bound, a natural first inclination is to show that among a sufficiently large set of directions in $\mathbb{R}^d$, there must exist correlated directions. In particular, it is possible to show that for d small relative to m , there exists $i \in [m]$ and k choices of j such that $\langle a_i, a_j \rangle > \frac{\epsilon}{k}$ and such that $\langle a_i, a_j \rangle$ have the same sign (and thus compound in interference). However, this approach does not generally work when $A \neq B$, since we do not strictly require representation directions of different features (or probe directions of different features) to be orthogonal. For $i \neq j$, it is only necessary for the inner product of feature i 's representation vector to be orthogonal to feature j 's representation vector. This means that there may exist "good" sets of representation directions and probe directions among which many are pairwise correlated.

In fact, in [sec:feature-geometry], we show that there are constructions with $B \neq A$ where probe directions are highly correlated with each other and representation directions are also highly correlated with each other. So while showing the existence of correlated vectors gives a lower bound in the case $A=B$, it does not suffice in general.

[Lower bound sketch.] It would suffice to show that if $\langle b_i, a_j \rangle$ is near 1 (as is required if $B^T A e_j \approx e_i$) and d is small, then there must exist many pairs $i \neq j$ such that $\langle b_i, a_j \rangle$ is large. In particular, we would like to find a feature i such that there are k choices of j such that $\langle b_i, a_j \rangle > \frac{\epsilon}{k}$ and such that the interference is all in the same direction. To do this, it suffices to find $2k$ such vectors a_j with $\langle b_i, a_j \rangle > \frac{\epsilon}{k}$ (by the pigeonhole principle).

We employ a few ideas. The first is to use a result of [alon2003problems], which demonstrates that low-rank matrices that are 1 on the diagonal must have a large off-diagonal entry. (This result also notably provides a near-optimal lower bound on the Johnson-Lindenstrauss Lemma.) In particular, we know that $B^T A$ has rank at most d , and we can minimally scale rows to allow diagonal entries to be 1. Note, however, that applying the result to $B^T A$ itself does not suffice since this only guarantees the existence of a single interfering pair of features.

Therefore, the second main idea is to apply the result instead to large principal submatrices of $B^T A$, which demonstrates that any submatrix of $B^T A$ must have a large off-diagonal entry. Equivalently, among any large subset $R \subseteq [m]$, there must exist a pair $i, j \in R$ with $i \neq j$ such that $\langle b_i, a_j \rangle$ is large.

Finally, we consider a graph reduction and apply Turan's theorem. Consider a graph G with vertices $[m]$ and such that there is an edge between i and j if and only if $\langle b_i, a_j \rangle > \frac{\epsilon}{k}$ or $\langle b_j, a_i \rangle > \frac{\epsilon}{k}$. Then G has no independent set of size r . Then we can apply Turan's theorem (which gives an upper bound on the number of edges in a graph without an r -clique) on $\overline{G}$, hence providing a lower bound on the number of edges in G , which in turns gives a lower bound on the number of pairs of interfering features. Then we can use the pigeonhole principle to show that there must exist some probe vector that has large interference with many representation vectors, as desired.

```
\begin{figure}
\centering
\begin{tikzpicture}[
  >=stealth,
  scale=2.5,
  vector/.style={->, very thick},
  axis/.style={very thin, gray!60},
  grid/.style={thin, gray!20}
]

\draw[grid] (-0.1,-0.8) grid (1.2,1.2);

\draw[axis,->] (-0.1,0) -- (1.2,0) node[right] {};
\draw[axis,->] (0,-0.8) -- (0,1.3) node[above] {};

\draw[vector,Bittersweet] (0,0) -- (0.5,0.866)
  node[above right] {$a_{\texttt{cat}} = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$};

\draw[vector,Bittersweet] (0,0) -- (1,0)
  node[below right] {$a_{\texttt{happy}} = (1,0)$};

\draw[vector,ForestGreen] (0,0) -- (0.1,1.547)
  node[above left] {$b_{\texttt{cat}} = \left(0, \frac{2}{\sqrt{3}}\right)$};

\draw[vector,ForestGreen] (0,0) -- (1,-0.577)
  node[below right] {$b_{\texttt{happy}} = \left(1, -\frac{1}{\sqrt{3}}\right)$};

\end{tikzpicture}
```

[Representation vectors $a_{\texttt{cat}}, a_{\texttt{happy}}$ and probe vectors $b_{\texttt{cat}}, b_{\texttt{happy}}$] that yield perfect linear recovery of the $\texttt{cat}$ and $\texttt{happy}$ features. Notice that while the representation vectors are not orthogonal, the probe vectors are able perfectly extract features since the $\texttt{cat}$ probe is orthogonal to the $\texttt{happy}$ representation and the $\texttt{happy}$ probe is orthogonal to the $\texttt{cat}$ representation. Indeed, observe that $b_{\texttt{cat}} \cdot a_{\texttt{cat}} = 1$ and $b_{\texttt{cat}} \cdot a_{\texttt{happy}} = 0$, and $b_{\texttt{happy}} \cdot a_{\texttt{cat}} = 0$ and $b_{\texttt{happy}} \cdot a_{\texttt{happy}} = 1$.

\end{figure}

\subsection{Contributions: Implications for the Linear Representation Hypothesis and the Theory of Language Models}

While linear representations have long been a central character in the empirical study of language models

→ \citep{mikolov2013distributed,alain2016understanding}, it has been unclear whether the LRH can provide a theoretical basis by which to study
→ language models. Here, we explore this possibility by providing mathematical foundations for the linear representation hypothesis. The framework we
→ introduce provides definitional clarity on the LRH, cleanly distinguishing the distinct assumptions of linear representation and linear accessibility, and
→ allowing us to formally consider the amount of superposition (i.e., the number of features stored relative to the number of neurons) possible under the
→ LRH.

The framework allows us to then derive theoretical implications of the linear representation hypothesis. Mathematically, we establish a quantitative gap
→ between compressed sensing and \textit{linear} compressed sensing. Our main results demonstrate dependencies between superposition (the
→ number of features stored relative to the number of neurons), sparsity (the number of features active per input), accessibility (the precision by which
→ the next layer can access features stored in the previous layer), and feature geometry (the arrangement of linear directions corresponding to
→ features).

Settled source reading Lower bound: add $\varepsilon < 1$ to the printed condition on ε .

Expanded PaperInterface specification

```
∀ {epsilon : ℝ},
  0 < epsilon →
  epsilon < 1 →
  GKP26LinearRepresentationHypothesis.linearCompressedSensingLowerSourceLogScale
  → =O[GKP26LinearRepresentationHypothesis.linearCompressedSensingLowerSourceFilter
    epsilon]
  fun (p : ℕ × ℕ) =>
    ↑(GKP26LinearRepresentationHypothesis.paper_linear_compressed_sensing_dimension p.1 p.2 epsilon)
```

Lean-elaborated claim roles

```
1. parameter: ℝ
2. assumption: 0 < epsilon
3. assumption: epsilon < 1
4. conclusion: GKP26LinearRepresentationHypothesis.linearCompressedSensingLowerSourceLogScale
→ =O[GKP26LinearRepresentationHypothesis.linearCompressedSensingLowerSourceFilter
  epsilon]
fun (p : ℕ × ℕ) => ↑(GKP26LinearRepresentationHypothesis.paper_linear_compressed_sensing_dimension p.1 p.2 epsilon)
```

Lean proof endpoint (built separately)

GKP26LinearRepresentationHypothesis.paper_lower_bound_dimension_isBigO_source_log_source_regime

Recorded source-to-Spec screening Verdict: matches

Reason: Compared the Lower Bound theorem at cited publication:220-227 and the supplied cited publication:327-380 context with the expanded target and all six supplied prerequisites, honoring LRH-LOWER-BOUND-NONDEGENERATE-EPSILON-01 only for this row. Epsilon is universally quantified outside the asymptotic assertion with $0 < \text{epsilon} < 1$, so its Big-O constant may depend on epsilon, as in $\Omega \text{ epsilon}$. For $p = (m,k)$, the displayed scale $k^2 \cdot (\log(m/k)/\log(k))$ is algebraically the printed $(k^2/\log(k)) \cdot \log(m/k)$. The filter takes the joint natural-size limit restricted to $2 \leq k$ and $5 \cdot k^3 < \text{epsilon}^2 \cdot m$; its epsilon bounds are already supplied by the public hypotheses. Under positive epsilon and $k \geq 2$, the squared strict inequality forces $m > 0$ and is equivalent to $\text{epsilon} > \sqrt{5} \cdot k^{(3/2)}/\sqrt{m}$. The $k \geq 2$ condition holds eventually in this limit and avoids $\log(k) = 0$. In the approved epsilon regime the inequality also gives $m > k$, making the displayed logarithmic scale positive; consequently scale $=O$ dimension expresses the stated lower bound, not an upper bound on dimension. The dimension wrappers display a positive-epsilon minimum-dimension choice and the predicate requiring feasibility at d and $d \leq$ every feasible d' ; the nonpositive-epsilon fallback is outside this target. The source proof sketch discusses unrestricted representation and probe matrices, and none of these supplied target dependencies imposes $A = B$ or an orthogonality restriction. This is an ordinary match under the approved epsilon clarification, not a corrected-target verdict.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 7

Source locator: cited publication, lines 412-418

Verbatim source input

```
\begin{lemma}[Incoherence of Random Matrices]\label{prop:incoherent}
  Let  $C \in \mathbb{R}^{d \times m}$  be a matrix with i.i.d. scaled Rademacher entries  $\pm \frac{1}{\sqrt{d}}$ . For
  \begin{equation}
    d = O\left(\frac{1}{\log m + \log \frac{1}{\delta}}\right) \mu^2,
  \end{equation}
   $C$  is  $\mu$ -incoherent with probability at least  $1 - \delta$ .
\end{lemma}
```

Expanded PaperInterface specification

```
∀ {m : ℕ} {mu delta : ℝ},
  2 ≤ m →
  0 < mu →
  0 < delta →
  delta ≤ 1 →
  ∃ (d : ℕ),
    0 < d ∧
    ↑d ≤ Real.log (2 * ↑m ^ 2 / delta) / (mu ^ 2 / 2) + 2 ∧
    (AppliedModelingLib.measureProb
      (AppliedModelingLib.Probability.RademacherMatrix.rowsMeasure (Fin m) (Fin d))
      fun (ω : Fin d → Fin m → Bool) =>
        ∀ {i j : Fin m},
          i ≠ j →
            |AppliedModelingLib.Math.LinearCompressedSensing.inner
              (AppliedModelingLib.Probability.RademacherMatrix.scaledMatrix ω i)
              (AppliedModelingLib.Probability.RademacherMatrix.scaledMatrix ω j)| <
              mu) ≥
            1 - delta ∧
    ∀ (ω : Fin d → Fin m → Bool) (i : Fin m),
      AppliedModelingLib.Math.LinearCompressedSensing.inner
        (AppliedModelingLib.Probability.RademacherMatrix.scaledMatrix ω i)
        (AppliedModelingLib.Probability.RademacherMatrix.scaledMatrix ω i) =
        1
```

Lean-elaborated claim roles

```
1. parameter: ℕ
2. parameter: ℝ
3. parameter: ℝ
4. assumption: 2 ≤ m
5. assumption: 0 < mu
6. assumption: 0 < delta
7. assumption: delta ≤ 1
8. conclusion: ∃ (d : ℕ),
  0 < d ∧
  ↑d ≤ Real.log (2 * ↑m ^ 2 / delta) / (mu ^ 2 / 2) + 2 ∧
  (AppliedModelingLib.measureProb (AppliedModelingLib.Probability.RademacherMatrix.rowsMeasure (Fin m) (Fin d))
    fun (ω : Fin d → Fin m → Bool) =>
      ∀ {i j : Fin m},
        i ≠ j →
          |AppliedModelingLib.Math.LinearCompressedSensing.inner
            (AppliedModelingLib.Probability.RademacherMatrix.scaledMatrix ω i)
            (AppliedModelingLib.Probability.RademacherMatrix.scaledMatrix ω j)| <
            mu) ≥
          1 - delta ∧
  ∀ (ω : Fin d → Fin m → Bool) (i : Fin m),
    AppliedModelingLib.Math.LinearCompressedSensing.inner
      (AppliedModelingLib.Probability.RademacherMatrix.scaledMatrix ω i)
      (AppliedModelingLib.Probability.RademacherMatrix.scaledMatrix ω i) =
      1
```

Lean proof endpoint (built separately)

GKP26LinearRepresentationHypothesis.paper_random_rademacher_strict_incoherence_source_rate

Recorded source-to-Spec screening Verdict: matches

Reason: Under approved context LRH-RANDOM-INCOHERENCE-NONDEGENERATE-RANGE-01, the source lemma at cited publication:412-418 is read for $m \geq 2$, $\mu > 0$, and $0 < \delta \leq 1$. The expanded target uses the i.i.d. scaled Rademacher row measure, gives an explicit d bounded by $\log(2*m^2/\delta)/(\mu^2/2)+2$, proves probability at least $1-\delta$ for strict off-diagonal incoherence, and records unit self-inner-products, matching the source rate and conclusion.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 8

Source locator: cited publication, lines 213-217

Verbatim source input

```
\begin{theorem}[Upper Bound]\label{thm:upper}
\begin{equation}
d(m,k,\epsilon) = O_{\epsilon}\left(k^2\log m\right).
\end{equation}
\end{theorem}
```

[Next verbatim source excerpt]

Taking $\mu = \frac{\epsilon}{k}$ and applying \Cref{prop:incoherent}, we have

```
\begin{equation}
d(m,k,\epsilon) = O\left(\frac{k^2\log m}{\epsilon^2}\right),
```

Expanded PaperInterface specification

```
∀ {epsilon : ℝ},
0 < epsilon →
(fun (p : ℕ × ℕ) =>
  ↑ (GKP26LinearRepresentationHypothesis.paper_linear_compressed_sensing_dimension p.1 p.2
    epsilon)) = O[Filter.atTop]
fun (p : ℕ × ℕ) => ↑ p.2 ^ 2 * Real.log ↑ p.1
```

Lean-elaborated claim roles

```
1. parameter: ℝ
2. assumption: 0 < epsilon
3. conclusion: (fun (p : ℕ × ℕ) =>
  ↑ (GKP26LinearRepresentationHypothesis.paper_linear_compressed_sensing_dimension p.1 p.2 epsilon)) = O[Filter.atTop]
fun (p : ℕ × ℕ) => ↑ p.2 ^ 2 * Real.log ↑ p.1
```

Lean proof endpoint (built separately)

GKP26LinearRepresentationHypothesis.paper_upper_bound_dimension_isBigO_k_sq_log_m

Recorded source-to-Spec screening Verdict: matches

Reason: The source theorem at cited publication:213-217 and its displayed construction at lines 450-452 state $d(m,k,\epsilon) = O_{\epsilon}(k^2 \log m)$. For each positive epsilon, the expanded target is exactly the minimum linear-compressed-sensing dimension IsBigO of $k^2 \log(m)$ at the product atTop filter, so the dependence, quantification, and conclusion agree.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 9

Source locator: cited publication, lines 436

Verbatim source input

In particular, the result guarantees the existence of a μ -incoherent matrix with $d = O(\frac{\log m}{\mu^2})$.

Expanded PaperInterface specification

$\forall \{m : \mathbb{N}\} \{ \mu : \mathbb{R} \},$
 $2 \leq m \rightarrow$
 $0 < \mu \rightarrow$
 $\exists (d : \mathbb{N}) (A : \text{GKP26LinearRepresentationHypothesis.paper_feature_matrix } m \ d),$
 $\uparrow d \leq 8 / \mu^2 * \text{Real.log } (2 * \uparrow m^2) + 2 \wedge$
 $\text{GKP26LinearRepresentationHypothesis.paper_strict_mu_incoherent } A \ \mu$

Lean-elaborated claim roles

- 1. parameter: $\mathbb{N}$
- 2. parameter: $\mathbb{R}$
- 3. assumption: $2 \leq m$
- 4. assumption: $0 < \mu$
- 5. conclusion: $\exists (d : \mathbb{N}) (A : \text{GKP26LinearRepresentationHypothesis.paper_feature_matrix } m \ d),$
 $\uparrow d \leq 8 / \mu^2 * \text{Real.log } (2 * \uparrow m^2) + 2 \wedge \text{GKP26LinearRepresentationHypothesis.paper_strict_mu_incoherent } A \ \mu$

Lean proof endpoint (built separately)

GKP26LinearRepresentationHypothesis.paper_rademacher_strict_incoherent_exists_source_rate

Recorded source-to-Spec screening Verdict: matches

Reason: The source prose at cited publication:436 asserts existence of a μ -incoherent d -by- m matrix with $d = O(\log m / \mu^2)$. For $m \geq 2$ and $\mu > 0$, the expanded target supplies such a matrix, with unit self-inner-products and strict off-diagonal inner products below μ , and the explicit bound $d \leq (8/\mu^2)*\log(2*m^2)+2$ has exactly the stated asymptotic rate.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 10

Source locator: cited publication, lines 700-733

Verbatim source input

Here, we prove that when feature representations are restricted to the Boolean cube $\{0,1\}^m$, $\Omega(k^2 \log(m/k))$ embedding dimensions are required to linearly separate embeddings with and without a feature. A corollary is that adding a bias and monotonic activation function does not increase the asymptotic capacity of embeddings.

```
\begin{theorem}\label{thm:threshold}
  Suppose that there exist  $A, B \in \mathbb{R}^{d \times m}$  and  $t \in \mathbb{R}^m$  such that for all  $i \in [m]$  and  $k$ -sparse  $z \in \{0,1\}^m$ ,
  \begin{equation}\label{eq:pineapple}
    (B^T A z)_i > t_i
  \end{equation}
  if and only if  $z_i = 1$ . Then for  $k < \sqrt{m}$ ,
  \begin{equation}
    d = \Omega\left(\frac{k^2}{\log k} \log \frac{m}{k}\right).
  \end{equation}
\end{theorem}
It's worth noting here that taking  $\epsilon < \frac{1}{2}$  in \Cref{thm:upper} provides a nearly-matching upper bound of  $\Omega(k^2 \log m)$ , since
 $\epsilon < \frac{1}{2}$  implies that  $(B^T A z)_i < \frac{1}{2}$  when  $z_i = 0$  and  $(B^T A z)_i > \frac{1}{2}$  when  $z_i = 1$ . Similarly,
\Cref{thm:threshold} implies \Cref{thm:lower} in the case  $\epsilon < \frac{1}{2}$ . However, the lower bound in \Cref{thm:lower} does not directly
imply \Cref{thm:threshold} since we do not require that  $\|B^T A z - z\|_\infty$  is small here, but rather that  $A z$  linearly separates positive and
negative examples for each feature. That being said, the proof is very similar to that of \Cref{thm:lower}, relying on the same overall strategy.

\begin{proof}
  Suppose that there existed  $A, B \in \mathbb{R}^{d \times m}$  and  $t \in \mathbb{R}^m$  such that for all  $i \in [m]$  and  $k$ -sparse  $z \in \{0,1\}^m$ ,  $(B^T A z)_i > t_i$ 
  if and only if  $z_i = 1$ . Then notice that we could scale the  $i$ -th column of  $B$  and  $\gamma_i$  by any constant positive factor to get another
  solution. In particular, we can obtain a solution where  $B^T A$  is  $1$  on the diagonal. Therefore, it suffices to show that for  $d \leq$ 
 $O(\frac{k^2}{\log k} \log \frac{m}{k})$ , there does not exist  $B, A, t$  satisfying \eqref{eq:pineapple} and such that  $(B^T A)_{ii} = 1$  for all  $i \in [m]$ .

  Then we have that  $t_i \leq 1$ , since  $(B^T A e_i)_i > t_i$ . Then, along the lines of the proof of \Cref{thm:lower}, it suffices to find  $i \in [m]$  and
 $T \subseteq [m]$  with  $|T| = 2k$  and
  \begin{equation}
    |\langle b_i, a_j \rangle| > \frac{1}{k}
  \end{equation}
  for all  $j \in T$ . Indeed, this would imply the existence of a pair  $i \in [m]$ ,  $T \subseteq [m]$  with  $|T| = k$ , such that
  \begin{equation}
    (B^T A z)_i > k \cdot \frac{1}{k} = 1 \geq t_i,
  \end{equation}
  giving a contradiction.

  We will again apply \Cref{lem:alon} to submatrices of  $C$  of size  $r$ . Indeed, \Cref{lem:alon} implies that there exists a constant  $\alpha > 0$  such
  that if  $|R| = r$  and
  \begin{equation}
    \text{rank}(C_R) < \alpha \frac{\log r}{(\frac{1}{k})^2 \log k} = \alpha \frac{k^2}{\log k} \log r,
  \end{equation}
  then  $C_R$  has an off-diagonal entry with absolute value at least  $\frac{1}{k}$ . Then we can proceed as in the proof of \Cref{thm:lower}, taking  $r =$ 
 $\frac{m}{4k+1}$  and applying Turan's theorem.
\end{proof}
```

Expanded PaperInterface specification

```
GKP26LinearRepresentationHypothesis.thresholdActivationLowerSourceLogScale
 $\hookrightarrow$   $= O[\text{GKP26LinearRepresentationHypothesis.thresholdActivationOriginalSourceFilter}]$ 
fun (p :  $\mathbb{N} \times \mathbb{N}$ ) =>  $\uparrow(\text{GKP26LinearRepresentationHypothesis.thresholdSeparationDimension p.1 p.2})$ 
```

Lean-elaborated claim roles

```
1. conclusion: GKP26LinearRepresentationHypothesis.thresholdActivationLowerSourceLogScale
 $\hookrightarrow$   $= O[\text{GKP26LinearRepresentationHypothesis.thresholdActivationOriginalSourceFilter}]$ 
fun (p :  $\mathbb{N} \times \mathbb{N}$ ) =>  $\uparrow(\text{GKP26LinearRepresentationHypothesis.thresholdSeparationDimension p.1 p.2})$ 
```

Lean proof endpoint (built separately)

GKP26LinearRepresentationHypothesis.paper_threshold_separation_dimension_isBigO_lower_source_log_original_regime

Recorded source-to-Spec screening Verdict: matches

Reason: The theorem at cited publication:700-733 gives the threshold-separation lower bound $k^2 \log(m/k) / \log(k)$ for $k < \sqrt{m}$. The expanded target expresses that Ω statement as the displayed source scale being IsBigO of the minimum threshold-separation dimension along the original-source-regime filter, and the supporting scale expands to exactly $k^2 * (\log(m/k) / \log(k))$.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 11

Source locator: cited publication, lines 734-750

Verbatim source input

The following corollary makes the simple observation that if a feature cannot be linearly separated, then it also cannot be separated by adding a bias and
↪ monotonic activation function.

```
\begin{corollary}\label{cor:activation-bias}
  Suppose that there exist  $A, W \in \mathbb{R}^{d \times m}$ ,  $b \in \mathbb{R}^m$ , and a monotonically increasing activation function  $\sigma: \mathbb{R} \rightarrow \mathbb{R}$ 
  ↪ such that for all  $i \in [m]$ ,  $\|k\|_2$ -sparse  $z \in \{0,1\}^m$ , and  $x = Az$ ,
  \begin{equation}\label{eq:coconut}
    \sigma(W^T x + b) > 0
  \end{equation}
  if and only if  $z_i = 1$ . (Here,  $\sigma$  is applied component-wise.) Then for  $k < \sqrt{m}$ ,
  \begin{equation}
    d = \Omega\left(\frac{k^2}{\log k} \log \frac{m}{k}\right).
  \end{equation}
\end{corollary}
\begin{proof}
  Consider any  $i \in [m]$  and  $z \in \{0,1\}^m$  such that  $z_i = 1$  and  $z'_i = 0$ . Then  $\sigma(W^T A z + b) > \sigma(W^T A z' + b)$ . This
  ↪ implies that  $(W^T A z)_i > (W^T A z')_i$ . Therefore, there exists a threshold  $t_i$  such that for all  $z \in \{0,1\}^m$ ,  $(W^T A z)_i > t_i$  if and only if
  ↪  $z_i = 1$ . The result follows by applying \Cref{thm:threshold}.
\end{proof}
```

The proof establishes that probes of the form $g_i(x) = \text{ReLU}(W^T x + b)$ implies the existence of a linear probe. In particular, this further establishes
↪ that for $\epsilon < \frac{1}{2}$, the bound in \Cref{thm:lower} cannot be lowered asymptotically by expanding the class of probes to allow a bias
↪ and activation function.

Expanded PaperInterface specification

```
GKP26LinearRepresentationHypothesis.thresholdActivationLowerSourceLogScale
↪ = O[GKP26LinearRepresentationHypothesis.thresholdActivationOriginalSourceFilter]
fun (p : ℕ × ℕ) => ↑(GKP26LinearRepresentationHypothesis.activationSeparationDimension p.1 p.2)
```

Lean-elaborated claim roles

```
1. conclusion: GKP26LinearRepresentationHypothesis.thresholdActivationLowerSourceLogScale
↪ = O[GKP26LinearRepresentationHypothesis.thresholdActivationOriginalSourceFilter]
fun (p : ℕ × ℕ) => ↑(GKP26LinearRepresentationHypothesis.activationSeparationDimension p.1 p.2)
```

Lean proof endpoint (built separately)

GKP26LinearRepresentationHypothesis.paper_activation_separation_dimension_isBigO_lower_source_log_original_regime

Recorded source-to-Spec screening Verdict: matches

Reason: The source corollary at cited publication:734-750 gives the $k < \sqrt{m}$ lower scale $k^2 \log(m/k) / \log(k)$ for activation/bias separation. The expanded target expresses that Omega claim as the displayed source scale being IsBigO of the minimum activation-separation dimension along the displayed original-source-regime filter; the supporting scale has exactly $k^2 * (\log(m/k) / \log(k))$.

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation